

Machine learning-assisted surrogate construction for full-core fuel performance analysis

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ABSTRACT

Accurately predicting the behavior of a nuclear reactor requires multiphysics simulation of coupled neutronics, thermal-hydraulics and fuel thermo-mechanics. Traditionally, fuel performance analysis is performed standalone as result of the cost induced by the full-core analysis. This work is therefore motivated to improve the computational efficiency of full-core modeling by constructing fast-running surrogate for fuel performance quantities of interest using machine learning. Rule-based models are used to account for the spatial-temporal-dependent operating histories, and expanded space-filling design is employed to ensure the appropriateness of the training set. Proposed methodologies are verified on both standard and high burnup PWR cores, achieving at least ten thousand times acceleration compared to FRAPCON with satisfying prediction accuracy. Current work lays foundation for tighter coupling of the fuel performance analysis into the core design optimization framework. It also sets stage for full-core fuel performance analysis with high-fidelity tools like BISON whose computational cost is more burdensome.

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1. Introduction

Nuclear reactor core design involves multiphysics simulations to meet the targeted safety standards, enhance the operational flexibility, and optimize the neutron economy hence fuel consumption. Realistic modeling of a reactor core requires coupling of a complete set of physical phenomena that impacts neutronics, thermal-hydraulics, and fuel thermo-mechanical behaviors at the engineering scale. Neutronics investigates the change in fuel composition and the spatial temporal distribution of fission power. Thermal-hydraulics considers the fluid mechanics of coolant and core-wide heat transfer. Fuel performance analysis depicts the thermo-mechanical behavior of fuel rods and assemblies as well as chemical processes that take place at the fuel pin level. These three fundamental physics are interdependent in the way that each one provides boundary conditions or essential feedbacks to another. Neutronics calculates the amount of gamma heating to the coolant, as well as power distribution that serves as input to the fuel performance analysis. Thermal-hydraulics estimates the fluid temperature and density which feeds back to neutronics, and the coolant heat transfer coefficients serve as boundary condition in fuel performance analysis. Fuel performance modeling

provides essential temperature and density feedback from the fuel rod to both neutronics and thermal-hydraulics.

Simulation capabilities of coupled core-level neutronics and thermal-hydraulics have reached a relatively mature stage with fast runtimes. Such coupled simulations are often augmented with simplified fuel performance feedback including average fuel temperature and thermal expansion of the solid structures (e.g. fuel/clad). Coupling of the fuel performance analysis, on the contrary, remains predominantly a one-way procedure. Output from the coupled neutronics-thermal-hydraulics modeling is normally used as input to the fuel performance modeling, and independent fuel performance calculations are performed for several limiting fuel pins. These limiting pins are typically selected by taking the pins with highest linear heat generation rate and burnup levels from the core-level reactor physics calculations. Recently, there has been more concentration on incorporating fuel performance analysis into the integrated multiphysics simulations of light water reactors (LWR), as demonstrated in the Virtual Environment for Reactor Applications (VERA) deployed by Consortium for Advanced Simulation of Light Water Reactors (CASL) program (Turner et al., 2016) (Stimpson et al., 2018). In addition to providing feedbacks to neutronics and thermal-hydraulics during reactor design, fuel performance analysis gives insight to potential failures risks of fuel rods and offers guidance on operational procedures. Thermo-mechanical fuel rod constraints must not be violated during the

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Nomenclature

CASL	Consortium for Advanced Simulation of Light Water Reactors	MSE	mean squared error
CDI	cumulative damage index	NN	(feed-forward) neural network
CV	cross validation	PCI	pellet-cladding interaction
DOE	design of experiments	PCMI	pellet-cladding mechanical interaction
GP	Gaussian process regression	PLS	partial least-square regression
IFBA	integral fuel burnable absorber	QoI	quantity of interest
LCOE	Levelized Cost of Energy (LCOE)	RBF	radial basis function
LHGR	linear heat generation rate	RF	random forest
LUT	look-up table	RL	reinforcement learning
LWR	light water reactor	RMSE	root mean squared error
MAE	mean absolute error	rRMSE	relative root-mean-square error
ML	machine learning	SCC	stress-corrosion cracking
MPS	missing pellet surface	TD	theoretical density
		XGB	extreme gradient boosted tree

entire reactor operation (and the dry storage stage for the spent fuel) to maintain the fuel integrity hence radioactivity. Accurate prediction of rod failure risks ensures sufficient safety margin for operation limits, reserves flexibility for operation, and enhances the neutron economy by allowing more aggressive power ramp rates.

Since only fuel performance of small fraction of the rods are typically analyzed, conservative factors are adopted to ensure that these peak rods are representative of the true margins to performance targets. The conservative peak-rod analyses can be undesirable for two reasons. First, the failure risks are typically overestimated, posing unnecessarily stringent requirement on the operation limits. Second, although less time-consuming, the peak-rod analyses do not necessarily reproduce the most hazardous fuel behavior. Some of the limiting fuel performance metrics are not linearly dependent on the power level, but proportional to the integration of power (burnup) or possibly depend on more complicated format of power, such as during load following. Particularly, if the industry moves to high burnup fuel (>62 MWd/kgU), the feedback of fuel performance to core reload fuel licensing limits will be stronger. For instance, as fuel burnup increases, the rod internal plenum pressure increases which is not a direct function of a peak power level. The full-core fuel performance analysis is therefore strongly motivated to more accurately account for the core-wide response.

Despite the fact that the total number of independent code runs can be reduced by exploiting the core symmetry and shuffling scheme, the full-core simulation still involves thousands of rod simulations and is thus time-consuming. The approximate total runtime for a specific core loading pattern typically falls onto half a day in serial with fuel performance analysis tools FRAPCON. With higher fidelity fuel performance tools like BISON, the total runtime can be as high as 10,000 to 30,000 core-hours on a high computing cluster for a single full-core analysis (Stimpson et al., 2018). In a typical core reload pattern optimization process, currently nuclear utilities use expert input and/or automated tools to evaluate hundreds to millions (with use of surrogates) of patterns. The objective function for selection of the best patterns is void of any detailed fuel performance for obvious issue of high computational cost. This motivates creation of fast running accurate surrogate models for fuel performance analysis to achieve a tighter coupling with current core simulators. Particularly, we plan to integrate the surrogate model in a recent physics-informed reinforcement learning (RL) optimization framework (Radaideh et al., 2021) (Martineau, 2020). The current constraints and objectives employed in this physics-informed RL framework accommodate only coupled neutronics-thermal-hydraulics response from a commercial

licensed code. The use of RL holds promise in significantly reducing the number of needed core pattern evaluations compared to the traditional optimization methods employed by nuclear industry (e.g. simulated annealing or genetic algorithm). Therefore, we plan to leverage the time savings by incorporating more physics to the RL framework to achieve more robust core reload patterns. The RL optimization process still requires tens of thousands of possible core pattern evaluations causing the runtime with even low-fidelity fuel performance tools becomes prohibitive. Therefore, surrogate construction is again motivated to significantly reduce the computational cost of the full-core simulation without much sacrifice of prediction accuracy.

The current work investigates useful techniques to construct fast-running surrogates for full-core fuel performance simulation with emphasis on two pragmatically meaningful tasks. The first effort is made to accurately predict the full-core behavior based on a reduced number of independent rod simulations for certain (known) core patterns. The second intention focuses on providing an efficient physics-informed full-core surrogate for copious unobserved core patterns. Runtime of such full-core surrogate should fall onto the timescale of seconds (at least a thousand times faster) to be capable of assisting the developed RL framework by coupling the fuel performance feedback into the objective functions and constraints.

Two categories of methodologies have been explored for surrogate construction: look-up tables (LUTs) as a quick empirical linear mapping, and machine-learning (ML) algorithms as alternative data-driven tools. LUTs, although extensively applied in industry for accurate prediction of average fuel temperature, prove inadequate in predicting some critical fuel performance quantities of interest (QoIs) that embed great nonlinearity. ML algorithms are therefore employed to capture the complicated nonlinear physical phenomena. Feature engineering during data preparation is the key to assure the success of ML algorithms, especially with the complicated spatial-temporal operating history. Rule-based model is investigated in this work for feature extraction and dimensionality reduction. Experimental design is equally important to ensure the appropriateness of the training data such that the entire data range is sufficiently covered by the training samples. For essential fuel performance QoIs, different types of supervised learning algorithms are applied and compared, including linear regression (partial least square regression, PLS), ensembling techniques (random forests, RF, and extreme gradient boosted trees, XGB), neural networks (NN) and kernel methods (gaussian processes regression, GP, and support vector regression, SVR).

This paper is therefore organized in the following order. Section 2 identifies a complete set of essential fuel performance

QoIs that translate to safety design limits during reactor operation. [Section 3](#) presents the full-core fuel performance analysis result for one specific fuel loading pattern from a prototypical PWR design. [Section 4](#) illustrates LUT as an empirical surrogate, revealing its suitability and inadequacy for multiple fuel performance QoIs. [Section 5](#) introduces the ML algorithms along with the feature engineering and experimental design techniques used for data preparation. [Section 6](#) compares the predictive performance of LUTs and ML full-core surrogates for essential fuel performance QoIs on three unobserved standard PWR cores. A combination of surrogates is selected based on combined consideration of predictive accuracy and time complexity. Applicability of the proposed methodology is also extended onto two high burnup PWR cores. Both applications show similar prediction accuracy and significant runtime acceleration. [Section 7](#) summarizes the major observations and how the current work sets stage for future works.

2. Essential QoIs in full-core fuel performance analysis

Fuel performance analysis simulates the thermo-mechanical behaviors of fuel rods under irradiation, subject to vital constraints for structural integrity of fuel rods. Constraints are placed on the following QoIs based on the potential fuel failure mechanisms in LWRs (USNRCO, 1981):

- 1) **Fuel temperature.** Fuel temperature involves feedbacks from multiple procedures. The temperature field promotes the thermal expansion of fuel pellets, causing decreased gap between the fuel and cladding, which in turn leads to lower fuel temperature especially at the beginning of life. High temperature hence high temperature gradient in the fuel drives the diffusion of gaseous fission products, leading to more release of fission gas into the plenum, deteriorating the gap thermal conductivity and increasing the plenum pressure, which further elevates the fuel temperature. In addition, fuel swelling is promoted at high temperature to accommodate the solid and gaseous fission products as the burnup accumulates, which accelerates the gap closure hence pellet-cladding contact. Moreover, despite that the fuel melting point is 2865 °C, average fuel should be operated below 1400 °C to limit excessive fission gas release. For these reasons, the maximum fuel temperature is selected as one figure of merit in the current work.
- 2) **Plenum pressure.** Internal pressure of the fuel rod rises initially due to elevated gas average temperature at reactor startup and reduced plenum volume caused by fuel expansion and cladding creep down, later due to release of fission gases to the rod void volume. Plenum pressure can also be a strong function of the initial fill gas pressure and presence of annular pellets at top and bottom of the fuel stack that provide additional free volume. In addition, a fraction of Westinghouse PWR rods are coated with integral fuel burnable absorber (IFBA) in PWR fuel assemblies. ¹⁰B in the IFBA coating burns out quickly in the first power cycle and its transmutation produces helium, contributing significantly to the rod internal pressure. Plenum pressure greater than the coolant pressure can cause tensile cladding stress hence cladding creep-out (outward deformation of the cladding tube). Once the cladding creep-out rate exceeds the fuel swelling rate, the pellet-cladding gap increases, reducing the gap conductivity, consequently increasing the fuel temperature. Increased fuel temperature leads to additional fission gas release, higher rod internal pressure, hence further cladding creep-out and larger gap. To prevent such adverse

thermal feedback, the plenum pressure should not rise above the coolant pressure to also avoid the cladding lift-off issue.

- 3) **Cladding oxidation.** Waterside oxidation of the Zircaloy cladding progresses steadily during normal operation and aggressively under accident scenarios in LWRs. Growth of the oxide layer (brittle ceramic ZrO₂) leads to not only clad thinning, but more importantly loss of clad ductility due to enhanced solubility of oxygen in the β -Zr phase at temperatures above 1200 °C. For such reason, the maximum oxidation layer thickness is another key parameter that should not exceed 17% of the as-fabricated clad thickness during accident conditions. The pre-accident oxidation thickness is therefore important to capture since it controls the margin to the accident limit.
- 4) **Cladding hydrogen pickup.** Absorption and diffusion of hydrogen into the zircaloy cladding occur in parallel with the corrosion process on the outer surface of the cladding. Hydrogen precipitates out as brittle hydride phase in zirconium-based alloys due to relatively low solubility at LWR operation temperatures (80 to 100 ppm) (Barrett and Bragg-Sitton, 2012). Uniformly distributed hydride causes gross embrittlement in the cladding, affecting its mechanical behavior. Localized hydride phase forming at the tip of a crack can result in crack extension (delayed hydride cracking) which has been a concern for the fuel structural integrity during transients at sufficiently high burnup as well as dry storage of the spent fuel. For accident conditions, NRC regulates that the total amount of hydrogen generated during reactor operation should not exceed 1% of the hydrogen generated hypothetically if all the cladding material (excluding the plenum) reactors with water (Barrett and Bragg-Sitton, 2012).
- 5) **Pellet-cladding interaction (PCI).** The CASL program identified cladding failure through PCI as one challenge that remains to be properly addressed (Pawlowski et al., 2014). Potential PCI failure places conservative restrictions on the operational power ramp rates as PCI failures typically occur during severe local power ramps following low operating power. Pure mechanical interaction (pellet-cladding mechanical interaction, PCMI) and stress corrosion cracking (SCC) are the two contributors to the PCI-induced failure.
 - a) PCMI challenges the cladding integrity with high stress that possibly tears the cladding apart. Among the three principal stresses in the cladding tube (thin-shell), the axial (or longitudinal) and radial stress are usually less than the hoop stress, which acts circumferential and perpendicular to the axis and radius of the cylinder wall. Therefore, the cladding hoop stress is usually investigated rather than the other stress components. The tensile hoop stress/strain can be viewed as a transformed identifier for occurrence of PCMI. Early in life the gap stays open, and the cladding deforms inward due to combined impact of irradiation creep and net compressive pressure (rod internal pressure lower than the coolant pressure). The cladding hoop stress therefore remains compressive (negative) during this stage. Later in life the pellet-cladding contacts due to fuel thermal expansion and cladding shrinkage, the fuel pellets impose outward force onto the cladding, leading to tensile (positive) hoop stress. Cladding is able to withstand much higher compression than tension due to the asymmetric yield strength in Zirconium alloys (Zhai et al., 2014). The maximum compressive hoop stress that cladding undergoes due to pressure difference typically falls

onto 50–70 MPa in LWRs, far below the compressive yield strength of Zircaloy. The maximum tensile hoop stress due to PCMI, on the contrary, can go up to as high as hundreds of MPa, posing risk to clad integrity. It is also worth mentioning that gap closure hence PCMI are usually delayed (or even do not occur) in IFBA rods compared to regular rods, with the cladding hoop stress being compressive for most of the time. Consequently, IFBA rods are generally less vulnerable to PCMI failure. In conclusion, accurate prediction of tensile hoop stress/strain matters more than the compressive component, especially for non-IFBA rods. For such reason, the maximum hoop stress and hoop strain are included into the essential QoIs as indicators for PCMI.

- b) Stress corrosion cracking (SCC) induced by PCMI and corrosive fission products is another principal failure mechanism. There are two mechanisms behind the SCC failures: PCI-SCC, where the pellet radial crack causes stress concentration at the cladding inner surface hence SCC failure; PCI-MPS, where SCC failure happens due to manufacturing-induced missing pellet surfaces (MPS) (Rashid et al., 2011). Cumulative damage index (CDI) model has been developed to compute the accumulated damage in the cladding, which translates into rod failure probability (Roberts et al., 1979). Failure risks are calculated based on CDI values for both PCI-SCC and PCI-MPS in the current work based on (Hales et al., 2016) (Joseph and Rashid, 2020). Details about the failure risk computation is described in **Appendix A**. Rods with failure risk higher than 0.5 are identified as PCI-vulnerable in the full-core PCI monitoring.

This work thereby considers the following eight QoIs as a complete set of essential fuel performance responses taken as the maximum values over the entire fuel rod life: (1) maximum fuel temperature, (2) maximum plenum pressure, (3) maximum oxidation thickness in the cladding, (4) maximum hydrogen concentration in the cladding, (5) maximum hoop stress in the cladding, (6) maximum hoop strain in the cladding, (7) rod failure risk due to PCI-SCC, and (8) rod failure risk due to PCI-MPS.

3. Full-core fuel performance analysis

A full-core fuel performance analysis for one specific fuel loading pattern is shown in this section. The full-core simulation is performed using NRC-licensed fuel performance analysis code FRAPCON 4.0, based on the output from CASMO-SIMULATE simulation. CASMO-SIMULATE provides core design parameters and fuel specifications (Table 3.1) as well as core-wide spatial-temporal distribution of power which serve as input to FRAPCON for fuel performance modeling. This work adopts equilibrium 15×15 PWR cores with a quarter symmetry for demonstration of methodologies. Fig. 1 shows one representative core loading pattern in a quarter core outputted by the mentioned RL framework. The exact reactor physics characteristics of the core performance is not relevant to this work and not included. The core exhibits typical power and burnup distribution of a 4-Loop Westinghouse PWR plant. Each assembly contains a 17×17 fuel rod array, with the number of IFBA rods in each assembly specified at the bottom. The core symmetry and shuffling scheme are exploited to reduce the total number of independent code runs (Deng et al., 2018), such that only 6864 independent FRAPCON runs are required to model the entire core which contains a total number of 50,952 fuel rods. A shutdown period of 15 days follows the end of each fuel cycle. At the beginning of each cycle, a simplified stepwise history is

Table 3.1

Core design parameters and fuel characteristics (note: the values are not identical to any of the commercial Westinghouse fuel products).

Parameter	Value
Fuel/cladding material	UO ₂ / Zirlo
Core thermal power	3645 MWth
Total number of fuel rods	50,952
Core average linear heat generation rate (LHGR)	19.70 kW/m
Coolant mass flux	3434.50 kg/(m ² s)
Coolant pressure	15.51 MPa
Coolant inlet temperature	565.7 K
Active fuel stack length	365.76 cm
Rod outer diameter	9.50 mm
Cladding thickness	0.5715 mm
Fuel-cladding diametric gap thickness	82.55 μ m
Annular pellet inner diameter (IFBA rods only)	3.9172 mm
Total length of the drilled section (IFBA rods only)	40 cm
Fuel density	95.7 % theoretical density (TD)
Fuel enrichment	4.8 %
Initial filled pressure for non-IFBA rods	2.41 MPa
Initial filled pressure for IFBA rods	0.7 MPa
Plenum length	17.5 cm
Thickness of ZrB ₂ coating on pellets (IFBA rods only)	7.0 μ m
Boron-10 enrichment in ZrB ₂ coating	37 %
Density of ZrB ₂ coating	90 %TD

assumed for power ramp up within a total time of 6 days. The linear heat generation rate (LHGR) rises linearly from zero to 30% full power level in the first 10 hours and holds for 50 hours, subsequently elevates to 80% of full power in the next 30 hours and remains there for 20 hours, then arrives at 90% power level in another 10 hours and stays for 10 hours, and finally reaches the 100% power level in the succeeding 14 hours. Two other distinguished cores that were randomly produced by the optimization algorithm are employed in this work as a representative of the typical core design constraints.

The full core distribution of multiple QoIs are presented in Fig. 2. Fig. 2a demonstrates the distribution of peak linear power over the entire in-reactor life for each rod. Fig. 2b shows the maximum rod-average burnup, verified between the fuel performance simulation and neutronics calculation. The maximum fuel temperature (peak fuel centerline temperature) of the whole core in Fig. 2c is roughly 1775 K. The highest plenum pressure is up to 15.34 MPa, marginally below the coolant pressure (15.51 MPa) as shown in Fig. 2d. Oxidation of the cladding is described in Fig. 2e, where the maximum oxide layer thickness reaches 43 μ m, equivalently 7.5% of the total cladding thickness. The core-wide distribution of clad hydrogen concentration in Fig. 2f is identical to that of the oxidation thickness, as both are physical description of the same waterside corrosion process. Fig. 2g and h illustrate the maximum hoop stress and maximum hoop strain respectively. The cladding experiences significant tensile stress/strain during the power ramp up following each shutdown period due to PCI, posing threat to structural integrity. The cladding damage can be estimated by the cumulative damage index (CDI) model (Rashid et al., 1989) (Hales et al., 2016) for PCI-SCC and PCI-MPS respectively in Fig. 2i and j. Calculated CDIs correspond to rod failure probabilities (**Appendix A**) as explained in Fig. 2k and l.

Fig. 2m to o show core-wide maximum fuel temperature, maximum plenum pressure and maximum cladding hoop stress as a function of time for all rods in the reactor. Red squares mark the fuel pins with peak linear heat generation rate (LHGR), blue diamond markers represent the most burnt pins, and the rest of the fuel pins in the reactor are denoted with minor black dots. Among these three QoIs, the most limiting responses occur with neither

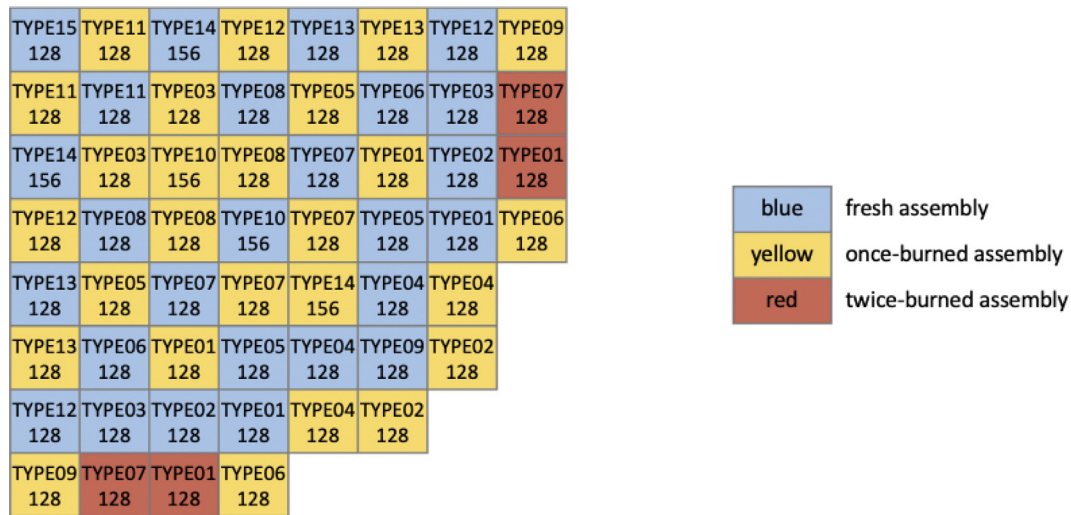


Fig. 1. Fuel loading pattern of a quarter core. Blue, yellow and red represent fresh, once-burned and twice-burned assemblies respectively. Number at the bottom denotes the number of IFBA rods in each assembly. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

peak rods nor the most burnt rods. This further proves the necessity to perform full-core analysis for accurate assessment of the safety margin, rather than the “peak rod” analysis which exaggerates the failure risk under conservative projections. It is worth pointing out that vast difference exists between regular rods and IFBA rods. Release of helium due to burnout of IFBA coating (ZrB_2) adds to the plenum pressure, impacts the gap conductance, delays the gap closure, and leads to a series of changes in corresponding mechanical responses. For such reason, two clear trends present in Fig. 2 (m) (n) (o).

4. Look-up table surrogate

Look-up tables (LUTs) have extensive application in nuclear engineering community as an alternative to physics-based models for material properties (Hagman and Reymann, 1979), heat transfer (Groeneveld et al., 2005), and flow regimes (El Nakla et al., 2013; Kirillov et al., 1991; Park and Roh, 1997) etc. On one hand, the data-driven LUTs are easy to implement and interpret. They may outperform the physics-based models in both prediction and efficiency, especially when understanding of the underlying physics is imperfect. On the other hand, LUTs are highly empirical and do not scale well to large input dimensionality especially when the interactions become important. Similar to most ML methods, they are typically designed within certain ranges hence fail to extrapolate beyond the ranges of validity.

In fuel performance analysis specifically, LUTs have been widely constructed for average fuel temperature prediction which is coupled to the neutronics analysis for temperature feedback (Stimpson et al., 2020). Such LUTs supply a two-dimensional mapping of the average fuel temperature values at discrete LHGR and burnup levels. The average fuel temperature at one specific LHGR and burnup can be easily accessed by two-dimensional interpolation of the LUT. Validity and accuracy of this LUT arise from the fact that the average fuel temperature calculation solely depends on LHGR as well as integral of local linear power with respect to core height and time (in other words, rod average burnup). The power history that variates with time has little impact onto the average fuel temperature due to the mathematical integration. For this reason, the LUT can be conveniently constructed by running independent FRAPCON simulations at different but constant LHGR levels. Practically, CASMO-SIMULATE simulates the core-wide power and burnup distribution, which can be directly utilized to calculate

the corresponding average fuel temperature during numerical iteration.

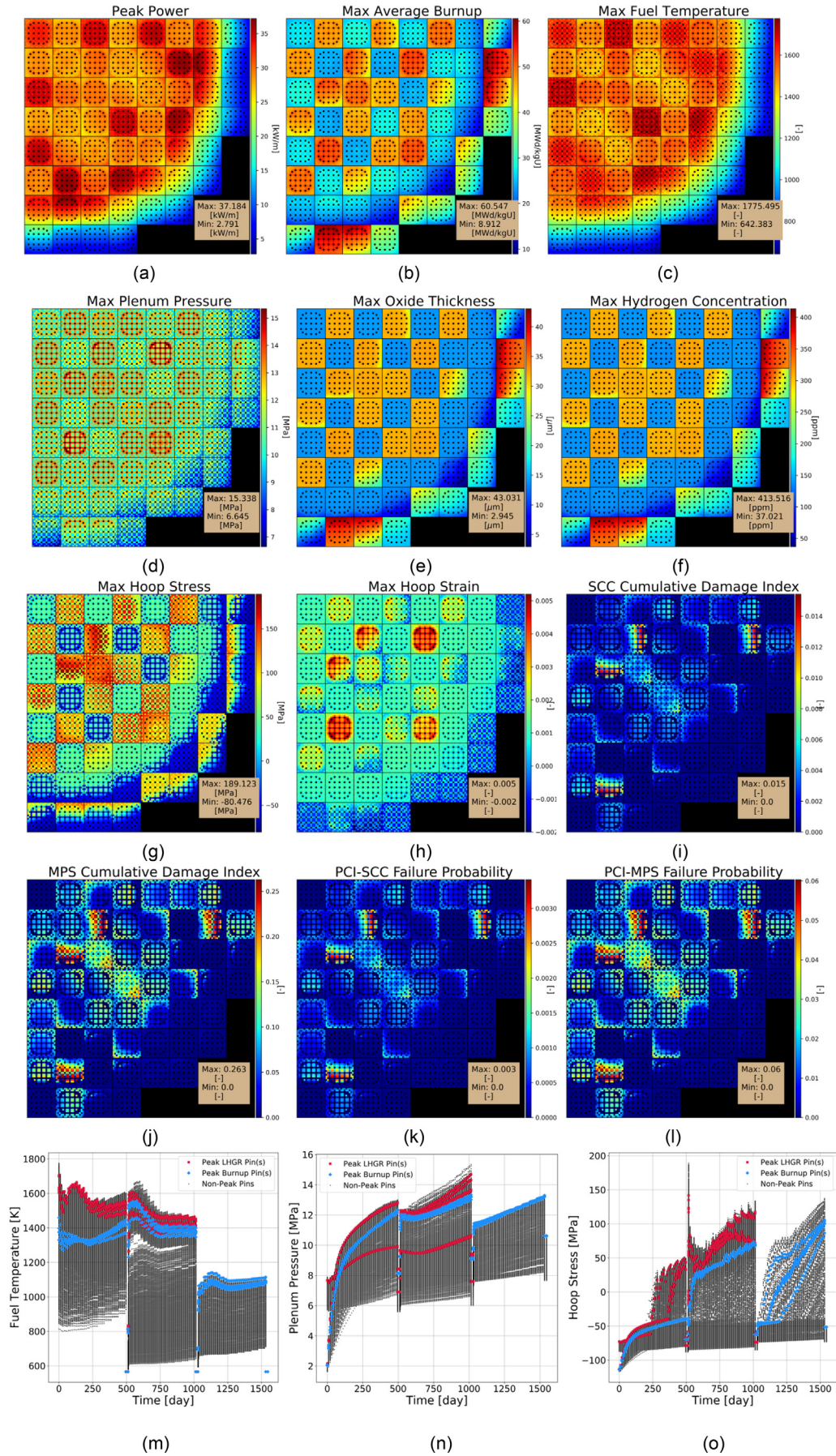
Despite the desirable performance with average fuel temperature, the suitability of LUTs does not necessarily extend to other fuel performance metrics mentioned in Section 2. As shown in Fig. 3, LUT predictions are plotted versus the true values, along with mean absolute error (MAE) and root mean squared error (RMSE). Predictivity coefficient (R^2) compares the variation in the residuals (difference between prediction and true values) to that of the observations. The model is considered more accurate with R^2 closer to 1.0, and practically models with $R^2 \geq 0.7$ are usually accepted as satisfactory (depending on the context and level of noise in data, sometimes R^2 of 0.5 is as high as achievable). Within this work's interest of investigation, LUTs perform the best for the maximum fuel temperature and the maximum plenum pressure. Forecasts for maximum hydrogen concentration and maximum oxide thickness are less accurate but still acceptable. Estimations for the essential PCI-related QoIs, on the contrary, are poor and require improvement.

Failure of LUTs originates from the complex physical phenomena these QoIs depend on. In fact, few QoI entirely relies on LHGR and average burnup as the average fuel temperature and the maximum fuel temperature do, but are more related to the time-dependent operating power. For instance, the maximum cladding hoop stress that serves as an indicator for PCI greatly depends on the occurrence of gap closure, which is closely related with the time-varying power “history” rather than one specific LHGR level. Such history-dependency comprises coupled thermo-mechanical responses much more complicated than simple integration of the spatial-temporal power, i.e., average fuel burnup. Although the power histories are usually distributed within certain finite ranges, in reality a reactor operation can follow countless possible operating histories. LUTs are inherently not designed for such tasks where the number of table entries are infinite.

5. Machine learning-assisted surrogate

5.1. Raw dataset

The reactor physics simulation by CASMO-SIMULATE generates core-wide spatial-temporal power history, which is used as FRAPCON input. The power history decomposes into two components:



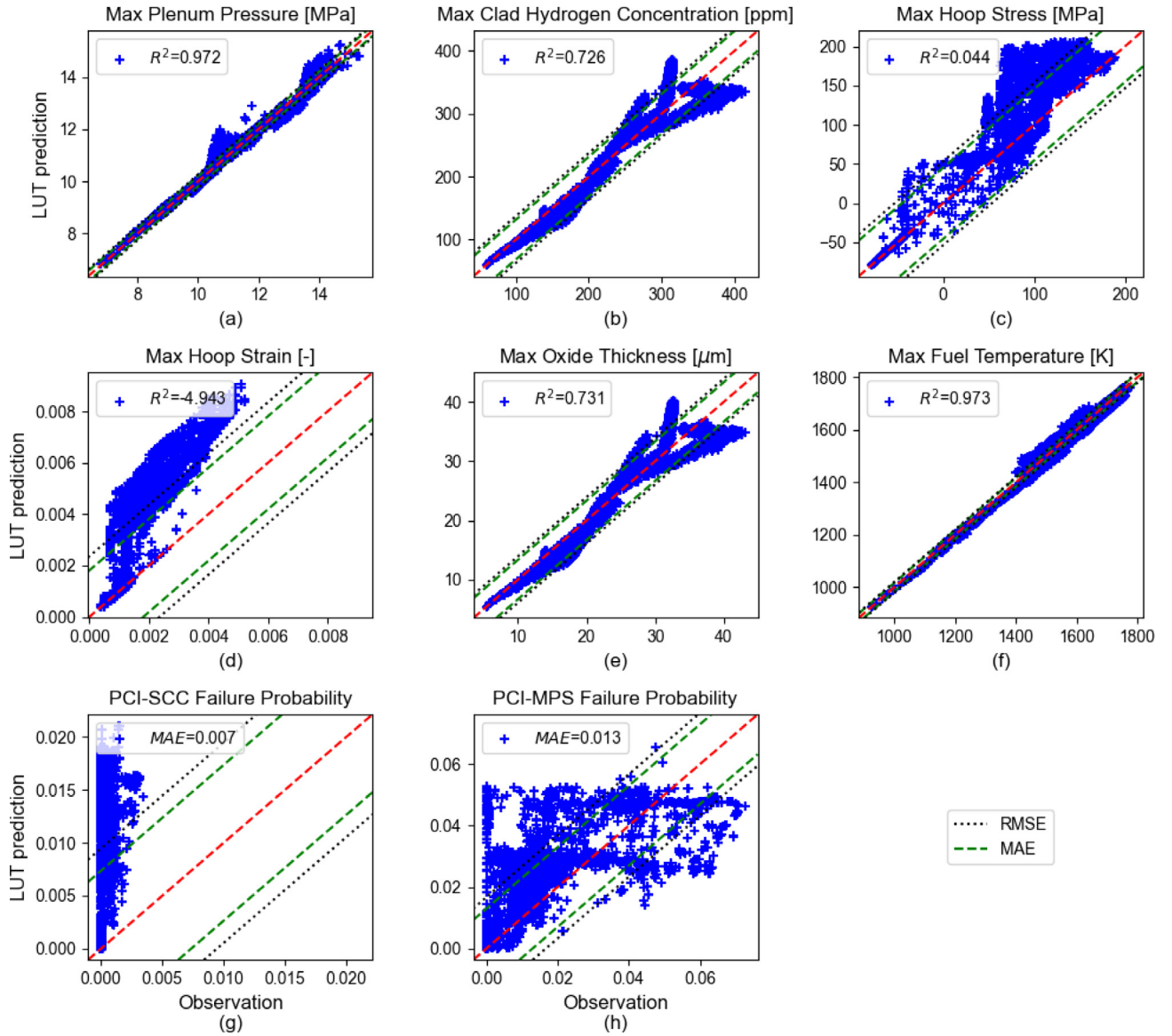


Fig. 3. Lookup table prediction of the fuel performance QoIs.

- (1). Linear heat generation rate (LHGR, averaged over core height) that varies with time, i.e., univariate time series that describe the cross-sectional power distribution.
- (2). Axial peaking factors that describe the axial power shape compared to LHGR at each timestep, i.e., multivariate time series that account for local power variations along the core height.

The current work investigates a 3-batch reactor core, where fuel assemblies undergo either one, two or three fuel cycles. Detailed analyses are presented in this paper only for the two-cycle rods for demonstration of methodology and brevity. The same methods are applied onto three-cycle rods and the results are directly sum-

marized in section 6.3. Fuel pins in different locations of the reactor core typically experience different power histories. Fig. 4a plots the LHGR histories for all fuel pins that go through two fuel cycles in the same core as shown in section 3. Fig. 4b shows the multivariate time series for axial peaking factor profiles. Each fuel rod is considered as one sample for ML that explores the relationship between spatial-temporal power history and fuel performance QoIs.

5.2. Feature engineering: rule-based model

There are multiple ways to perform feature extraction for the same physics problem. Appropriate raw features are those that not only correlate well with the output (good representation of

Fig. 2. Full-core fuel performance modeling result for a representative PWR core. Subfigures (a) to (i) exhibits the core-wide distribution of the maximum values of quantities of interest during the entire reactor operation. Subplots (m) to (o) show the evolution of maximum fuel temperature, plenum pressure and maximum hoop stress as a function of time. Fuel pins with peak linear power are marked with red squares, and the blue diamonds mark the most burnt fuel pins. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

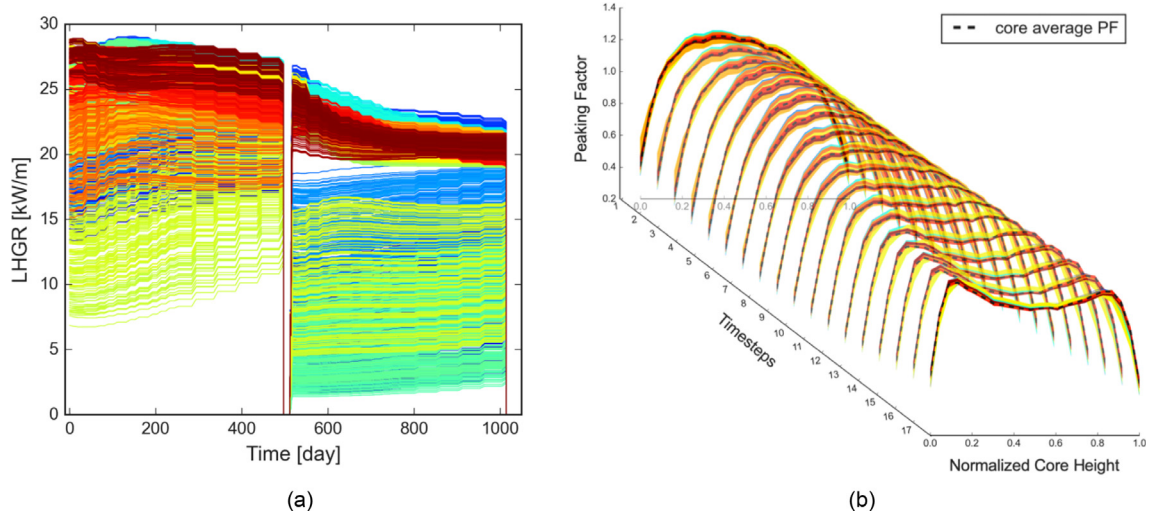


Fig. 4. Power histories for all pins that go through 2 fuel cycles in the same core. (a) LHGR time-series. (b) evolution of axial peaking factor (PF) profiles with time. The black dashed lines denote the core-average PF profile at each time step. Fuel rods are distinguished by color.

the input space), but also with moderate dimensionality. This work illustrates rule-based modeling as an effective approach to deal with the spatial-temporal complexity of power histories.

The rule-based model representation uses a set of rules to indirectly specify a mathematical model. When the rule-set is highly simpler than the original model it implies, such representation can be especially effective. This work utilizes polynomial regression for purpose of dimensionality reduction. The multivariate time series of PFs is simplified into a univariate time series by taking the maximum axial PF at each timestep to preserve the most limiting fuel performance QoIs. The axial PF profiles are of similar shapes to the core-average PF at each timestep (as shown in Fig. 4b), such that the maximum PF values become a valid representative of the axial profile. The two univariate time series (LHGR and maximum PF) are subsequently fitted by 4th-order polynomial regression for each power cycle (reactor shutdown and power startup periods removed). Fig. 5 shows a schematic of the fitted polynomials for one power cycle of a specific rod. The corresponding polynomial coefficients are considered as representative features of the power history. Additionally, rod type is encoded as a categorical variable with one-hot encoding for IFBA rods and non-IFBA rods. The feature space obtained in this way is thereby 21-dimensional for two-cycle rods.

In addition to providing an effective feature set, the rule-based model representation inherently comes with sufficient flexibility hence extensibility. This is critically important as the training data is expected to be designed with good space-filling property within ranges that are physically possible. While the available data at hand is usually limited, the polynomial representation allows creation of synthetic dataset into unseen but physically relevant regions with assistance of computational tools.

5.3. Experimental design: extended space-filling design

In the context of surrogate modeling, data collection is typically achieved through computer experiments (or code runs, sampling). Improving the space-filling property not only promotes computational efficiency in sampling, but also ensures appropriateness of the training data such that extrapolation can be avoided during generalization. The procedure to come up with an optimal design is often referred to as experimental design, or design of experiments (DOE). In this work, a synthetic training set is designed to sufficiently cover the extended feature space that corresponds to all possible power histories with physical desirability, to make sure that the surrogate is extensible to unseen core patterns. The experimental design is carried out by an extended space-filling design

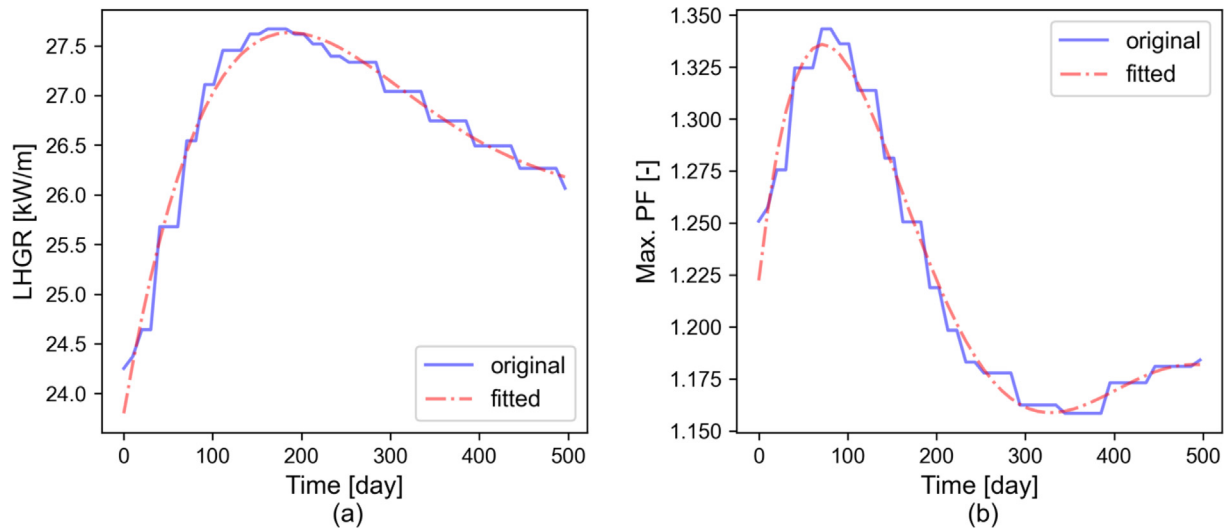


Fig. 5. Demonstrative polynomial regression in comparison to the original data in one power cycle of a typical rod, for (a) LHGR time series, and (b) maximum PF time series.

with the assistance of clustering and Latin Hypercube sampling (LHS).

For one specific fuel loading pattern, samples fall into certain ranges in the transformed feature space. Such ranges can be very dissimilar for different fuel loading patterns. This work takes three representative fuel loading patterns, and the feasible physical bounds (to the best of existing knowledge) are defined as the union of the outermost boundaries from the three available cores combined. No effort is made to retrieve the exact pairwise correlation between raw feature for two reasons. First, it becomes a challenging task as dimensionality increases. Second, it is unnecessary to retain all samples within the exact bounds. On the contrary, it is more beneficial to extend beyond the boundaries to accommodate for unobserved but physically desirable samples in the unseen cores patterns. The experimental design for such purpose is carried out following **Algorithm 1**. Details of the experimental design is illustrated and visualized step by step in **Appendix B**.

Algorithm 1: Extended space-filling design

- 1: Combine all available data in the feature space for several representative cores.
 - 2: Divide feature space into multiple subsets (clusters) $\Xi_i, i \in \{1, 2, \dots, m\}$, such that the combined samples are relatively uniformly distributed within each region.
 - 3: **For** each subdomain Ξ_i :
 Fit marginal probability distributions to each feature variable.
 Perform maximin Latin Hypercube sampling (LHS) according to the fitted marginal distributions. Denote the LHS sample set as Φ_i .
 - 4 Take union of the LHS sample sets $\Phi = \bigcup_{i=1}^m \Phi_i$
 - 5 Discard physically undesirable samples from Φ
-

LHS samples drawn in this way neglect the dependency between feature variables, whereby advantageously extend into unexplored regions with nice space-filling property. The outermost bounds are obtained based on available data to the best of existing knowledge, which can be consistently updated upon availability of more core patterns. The physical desirability in this work takes the principle of retaining the LHGR level between 0 and 30 kW/m, and the maximum PF value no greater than 1.6. Power histories are recon-

structed from drawn samples in the transformed feature space, and independent fuel performance analyses are ran at designed locations to constitute the synthetic ML training set.

5.4. ML algorithms

Supervised learning algorithms of different categories have been selected and compared to capture the fuel performance QoIs. Predictive performance of representative algorithms from linear regression, kernel method, neural network, and ensemble methods are investigated.

5.4.1. Partial least squares regression (PLS)

PLS (a.k.a. projection to latent structures) is an efficient regression method built upon covariance. In PLS, predictors are reduced to a smaller set of latent factors (orthogonal) that maximize the explained variance in dependent variables. Least squares regression is then performed between the latent factors and response variables. PLS can be especially useful when the predictors are correlated or when the number of predictors is similar to/higher than observations (Tobias, 1995).

5.4.2. Gaussian process regression (GP)

GP is introduced into this work as a representative of the kernel method. GP is a nonparametric and probabilistic model that calculates the probability of distribution over all permissible functions that fit the data. As a Bayesian method, a Gaussian process prior needs to be specified in the function space. The form of mean function and covariance function (kernel) in the Gaussian process prior is tuned based on the training data. The posterior distribution is calculated based on the Bayes rule, and the predictive posterior distribution can be obtained on points of interest. The merit of employing GP is that it naturally comes with uncertainty measurement in the predictive posterior distribution, which benefits the uncertainty quantification procedure in many engineering applications (Rasmussen, 2003). Nonetheless, the computational cost of training a GP explodes with increased input dimensionality or training size.

5.4.3. Support-vector regression (SVR)

SVR is another kernel method that is extended from support-vector machines to solve regression problems. The main idea is to first transform inputs to higher dimensional feature space via

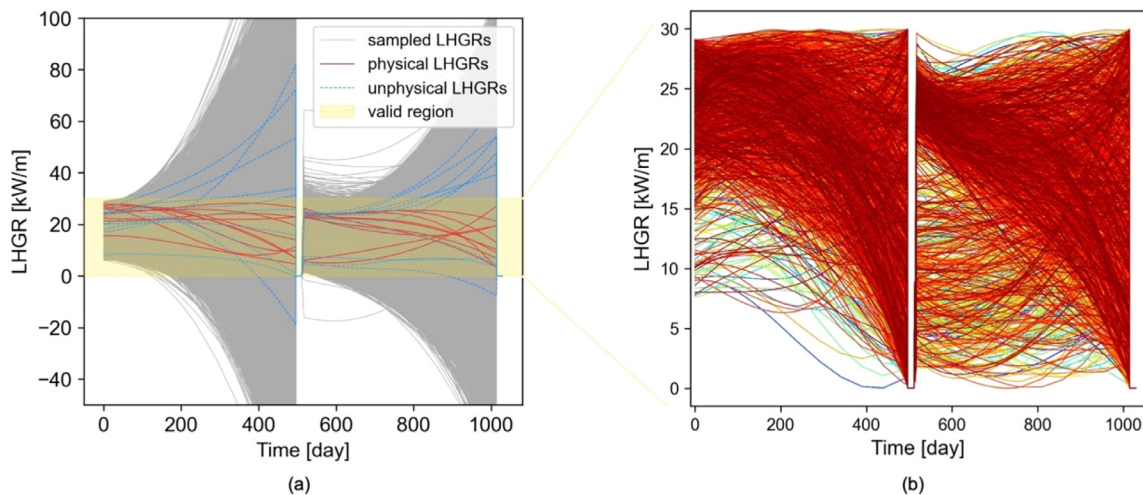


Fig. 6. Reconstructed power histories in the extended space-filling design. In (a), gray lines represent all sampled LHGRs. Red/blue lines are a few representatives of the physical/unphysical LHGR samples, where physical LHGR time series are those falling into the “valid region” (between 0 and 30 kW/m). All physical LHGR time series that are used as training set in this work in shown in (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

kernels so that the data becomes separable. A simple linear mapping can then be implemented in the transformed space for regression. Various kernels can be used, among which the radial basis function (RBF) kernel and polynomial kernel are the most powerful. Key advantages of SVR are its effectiveness especially in high dimensional spaces as well as the universal approximation capability (Hammer and Gersmann, 2003). SVRs are more efficient than GPs in general as only a subset of training data is used for support vectors. In the current work, SVR with RBF kernel is employed as an alternative kernel method to GP.

5.4.4. Neural network (NN)

As a universal approximator for arbitrary nonlinearity, feed-forward neural network (a.k.a. multilayer perceptron) is a promising ML method (Jain et al., 1996). NNs contains layers of interconnected nodes (neurons). Each neuron feeds the weighted sum of all inputs from previous layer to a nonlinear activation function. Hidden layers (intermediate layers between the input and output layer) extrapolate the salient features in the input data that embed predictive power, similar to the effect of feature engineering. Weights and biases are fine-tuned through iterative backpropagation to achieve minimal prediction error on the training data. Under most circumstances, NNs with one, two or three hidden layers provide sufficient predictive power of input–output nonlinearities. Three NN structures are investigated thereby in this work: single-layered NN (one hidden layer), two-layered NN (two hidden layers) and three-layered NN (three hidden layers).

5.4.5. Random forest (RF)

RF is a popular tree-based ensemble learning algorithm for its simplicity, efficiency and robustness. Individual decision trees are trained in parallel on different subsets of training data based on unique combinations of features (Louppe, 2014). Results from these decision trees are aggregated together for prediction with reduced variance, so RFs usually outperform decision trees and exhibit good generalization. RFs are easy to tune, robust to noises and training sample selection, and unlike other distance-based ML algorithms, do not require feature scaling. In the context of fuel performance modeling, many physical processes are triggered

above certain threshold, so the tree-based algorithms are expected to more accurately capture such discontinuities.

5.4.6. Gradient boosted tree (XGB)

Gradient boosted tree is another tree-based ensemble method that is gaining wide use. The key idea of “boosting” is to iteratively combine “weak learners” into one single “strong learner”. Functional gradient descent algorithm is used to optimize the cost function over the function space. Similar to RFs, boosted trees use an ensemble of decision trees for prediction, and feature scaling is not necessary. Unlike RF which combines a forest of random trees in parallel, decision trees in boosted trees are trained successively on the errors of previous trees. A series of shallow trees are ensemble in a nonrandom way, and the prediction error becomes smaller as more trees are added. Making predictions with boosted trees is fast and not memory-consuming once the model is built. Python package XGBoost is employed to implement the gradient boosted tree, and it is referred to as XGB (extreme gradient boosting) in this work (Chen and Guestrin, 2016).

6. Results and discussion

ML algorithms are implemented in Keras (Chollet, 2015), Scikit-learn (Pedregosa et al., 2011), and XGBoost (Chen and Guestrin, 2016) in Python 3.6. The dataset is partitioned into a training set (on which the ML model is trained) and a test set (on which the predictive performance of the model is evaluated). k -fold cross-validation is used for model selection. The training set is randomly shuffled prior to cross-validation, and is averagely split into k folds. Each one of the k folds is held out once to evaluate the performance of model trained on the rest ($k - 1$) folds. $k = 5$ is used in the current work. Mean and standard deviation of the model evaluation scores are stored as the assessment of model effectiveness. The test set is a holdout data set that has never been used for training, and the predictive performance of the trained model is evaluated on the test set. This work uses a synthetic training set from the experimental design, and evaluates the performance of surrogates on the real data from three PWR cores unobserved by the training set. This section presents the predictive performance of ML surrogates in comparison to LUT, selects the best combination of surrogates

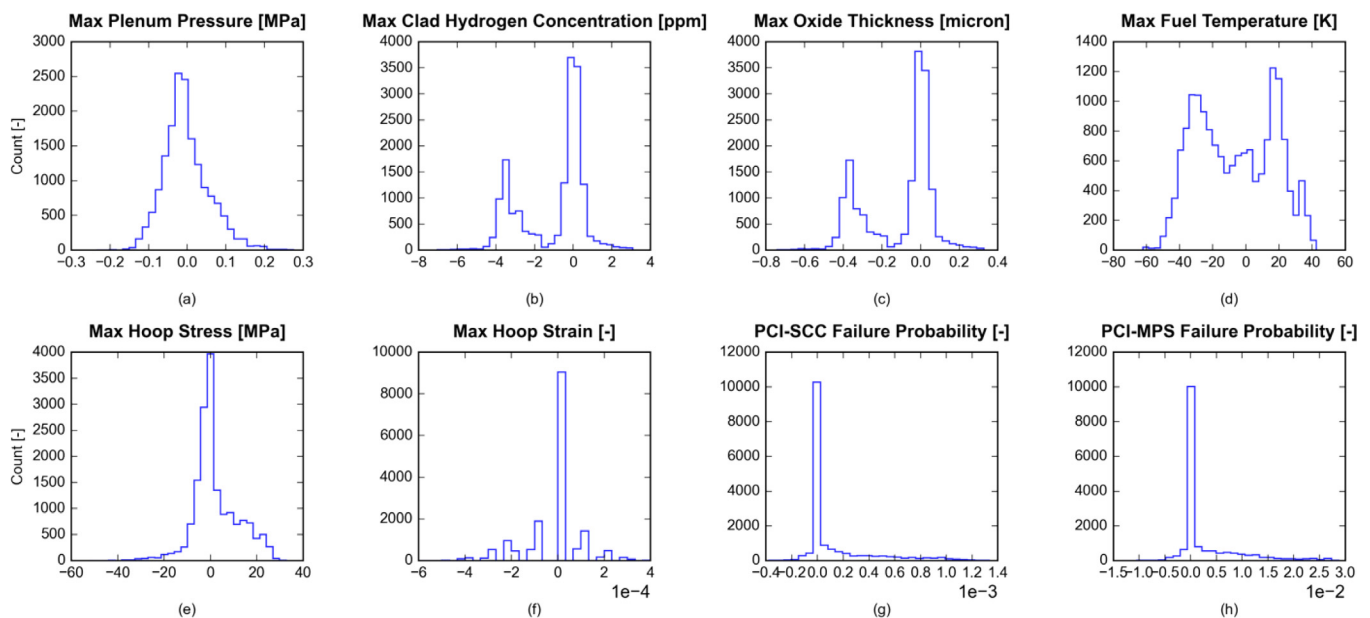


Fig. 7. Reconstruction errors for the eight fuel performance QoIs.

based on runtime analysis, and demonstrates the applicability of the developed methodologies on high burnup cores.

6.1. Predictive performance of ML surrogates

In this work, a total number of 9128 samples are drawn from the extended space-filling design as training set. Similarly, five-fold cross-validation is performed to select the best model structure. A separate ML surrogate is trained for each QoI with different ML algorithms. The predictive performance is tested on a total number of 16,632 realistic fuel pins from three distinguished cores.

The training set in this case is purely synthetic from experimental design on extended data range to accommodate unobserved cores. LHGR time series are reconstructed from polynomial coefficients in the feature space, as shown in Fig. 6. Although rods in real reactors may never experience some of the power histories in

Fig. 6, adding training samples at such locations help prevent extrapolation of the ML algorithms when predicting unexplored core patterns. Core-average PFs are used for PF profiles to preserve the location of maximum peaking factors. FRAPCON inputs are generated and ran based on the reconstructed operating conditions for training data collection.

Despite superior flexibility, the spatial-temporal operating condition reconstructed in this way inevitably loses details. Fig. 7 shows the reconstruction errors for the eight QoIs on the three available cores with a total number of 16,632 fuel pins. Such reconstruction errors result from a compromise between dimensionality of the feature space and reconstruction precision on the spatial-temporal dependent PF profiles. Presently the PF profile at each timestep is represented with the maximum PF value. More raw features can be employed for more accurate reconstruction of the PF profiles, while the improvement in reconstruction accuracy

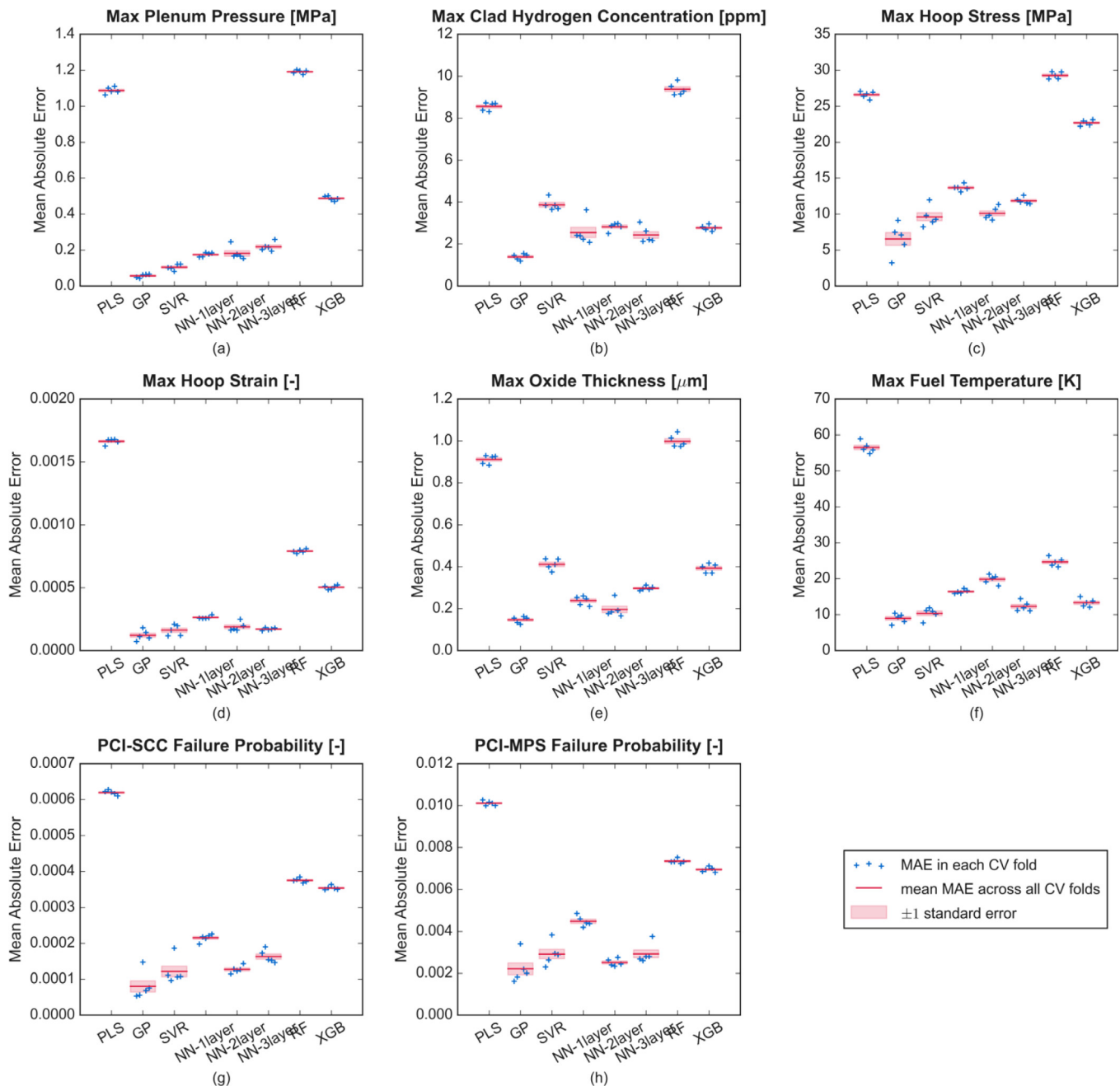


Fig. 8. Mean absolute predictive error on each hold-out fold during cross-validation.

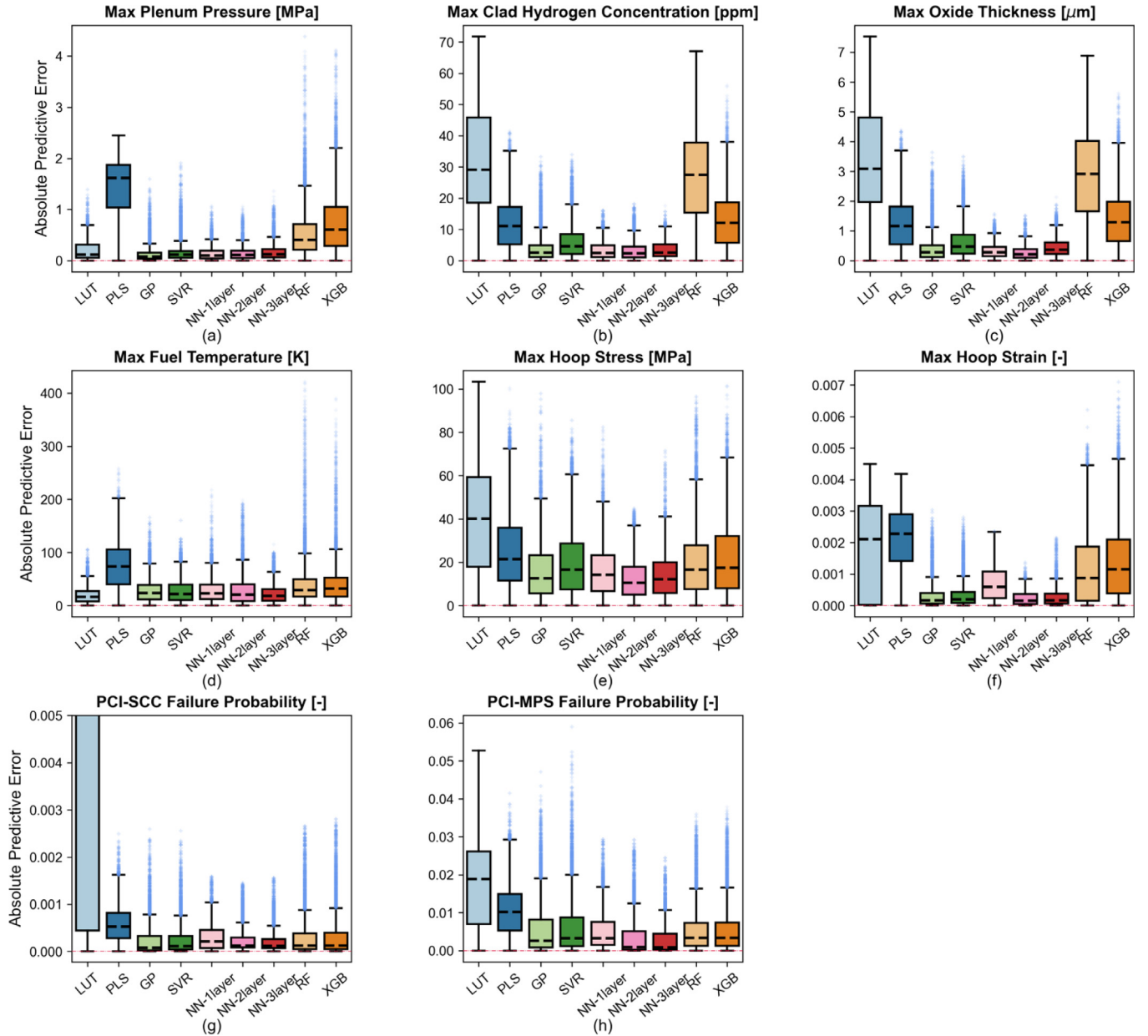


Fig. 9. Boxplots of absolute predictive error with LUT and ML algorithms on the three unseen cores for the eight fuel performance QoIs.

does not compensate for the loss of accuracy in ML training due to increased input dimensionality. Such reconstruction error is a necessary sacrifice to make and serves as a primary source of prediction error in the ML surrogates, as confirmed in Figure C.1 in **Appendix C** which compares the surrogate predictive error versus the reconstruction error of the best selected surrogate.

Fig. 8 shows the cross-validation averaged hold-out set performance. During model selection, the k -fold cross-validation is performed in the way that the ML model is trained on $(k - 1)$ folds and validated on the left (hold-out) fold. The predictive MAE on each one of the k hold-out folds are plotted as blue markers in Fig. 8. The solid lines represent the mean MAE across all hold-out folds. The shaded bars show the plus or minus one standard error as a measure of the 68% confidence interval. Ideally, the variation across folds should be small, indicating consistent mean absolute predictive errors across the k hold-out folds and therefore good generalization capability onto new data. Given similar cross-validation averaged hold-out set performance, models with simpler structures are favored over those with more complexity in

order to prevent overfitting. Most of the ML algorithms provide consistent performance (with small one standard error) on the hold-out sets. A few exceptions are the kernel methods especially for the failure probabilities, where the average estimate of MAE across the k folds is subject to relatively larger uncertainty due to random split of the cross-validation folds.

The predictive capability of aforementioned ML algorithms is tested on the three PWR cores (test set) that have no overlap with the training set, and is compared against LUT for each fuel performance QoI in Fig. 9. NNs and GP excel among the ML algorithms, while the performance of LUT only becomes acceptable for the maximum plenum pressure and the maximum fuel temperature. As an alternative visualization of the predictive performance, Fig. 10 (a) to (f) plots the cumulative data fraction versus relative absolute error, with the relative root-mean-square error (rRMSE¹) shown in the side table. The cumulative data fraction curves rise fas-

¹ Relative root-mean-square error (rRMSE) is defined as $rRMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{y}_i - y_i}{y_i} \right)^2}$ where y_i and $\hat{y}_i$ are the observation and surrogate prediction.

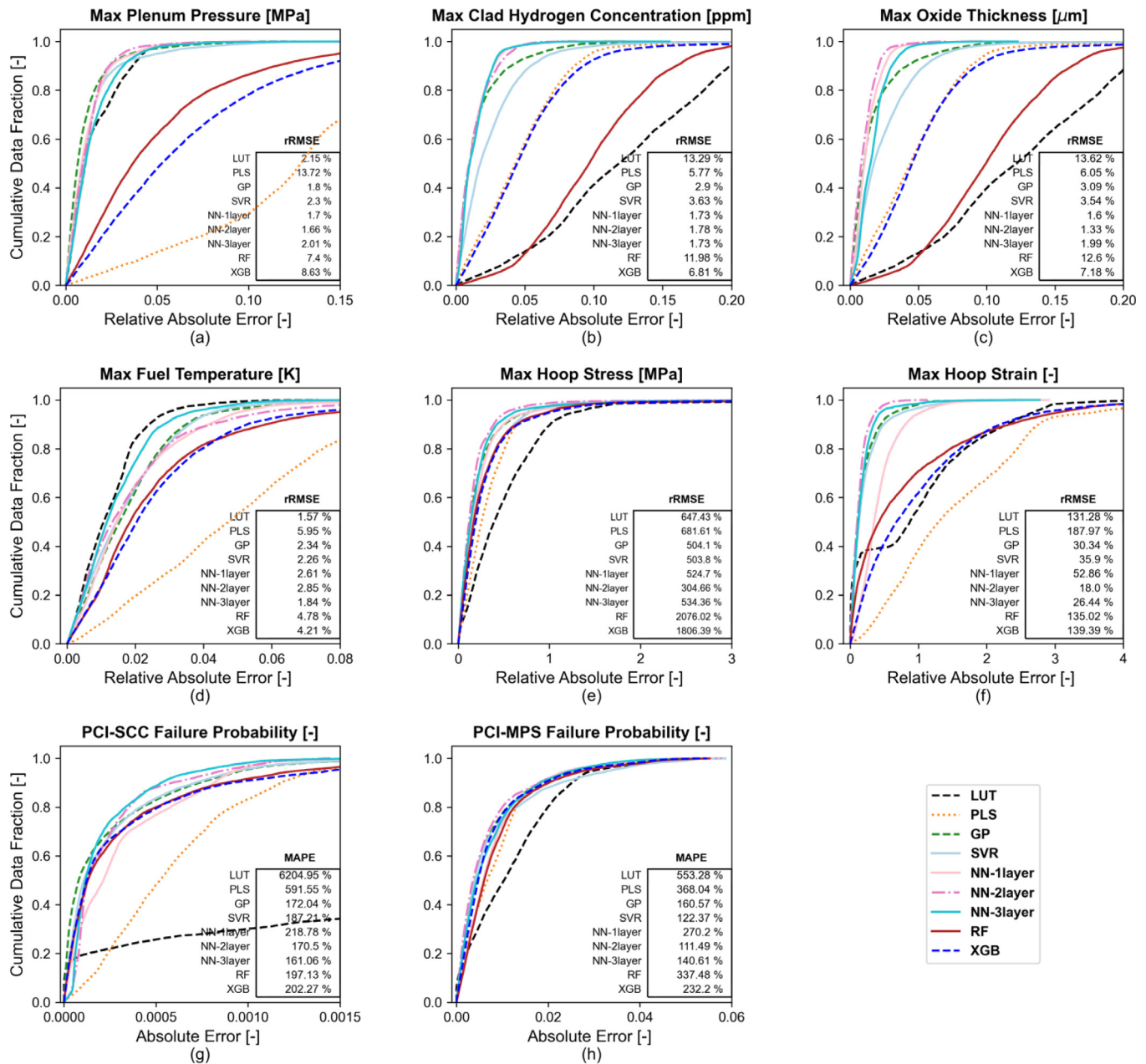


Fig. 10. Cumulative data fraction of (relative) absolute errors for LUT and ML algorithms for the eight fuel performance QoIs.

ter for surrogates with the better performance, at the same time subject to smaller rRMSE. The rod failure probabilities in Fig. 10 (g) and (h) are by definition small fractional numbers, under which case the absolute error and correspondingly mean absolute percentage error (MAPE) are used as more sensible error metrics. All subplots in Fig. 10 are horizontally zoomed in for better visibility.

A more detailed comparison of the surrogate performance is summarized in Table C.1 in Appendix C, including the predictivity coefficient R^2 , RMSE, MAE, maximum and minimum prediction error on the test set. R^2 compares the variation in the predictive error to that of the observations. RMSE and MAE are both calculated in this work, with MAE intended for average predictive error, and RMSE accounting for variance of the predictive error². RMSE is more sensitive to outliers than MAE due to quadratic scoring, and

the difference in the model rankings between the two metrics indicates presence of outliers. In addition, the maximum and minimum predictive errors are also extracted for outlier identification.

The best R^2 for the maximum plenum pressure, maximum hydrogen concentration, maximum oxide thickness, and maximum fuel temperature are all able to reach 0.95, indicating excellent predictivity. The maximum hoop stress is subject to greater prediction error due to the inherent uncertainty of its occurrence as well as relatively larger reconstruction error, while the best R^2 is still above 0.92. For some rods that go through high LHGR, the maximum cladding hoop stress occurs in the form of a spike at the beginning of the second power cycle due to PCI, leading to large tensile (positive) hoop stress. PCI may never take place in some other rods, so the maximum hoop stress remains compressive (negative) throughout the life. It is possible that other rods undergo relatively moderate PCI, such that the maximum hoop

² RMSE is always greater equal than MAE by definition.

Table 6.1
Time complexity for the surrogate

Surrogate	Training Complexity	Prediction Complexity	Nomenclature
LUT	–	$\mathcal{O}(1)$	
PLS	$\mathcal{O}(nd^2)$	$\mathcal{O}(d)$	
SVR	between $\mathcal{O}(n^2p)$ and $\mathcal{O}(n^3p)$	$\mathcal{O}(n_{sv}p)$	p : number of features n_{sv} : number of support vectors
GP (Raykar and Duraiswami, 2007)	$\mathcal{O}(n^3)$	$\mathcal{O}(n)$	
NN	–	$\Theta(dn_{l_1} + n_{l_1}n_{l_2} + \dots)$	n_{l_i} : number of neurons in i -th hidden layer
RF (Louppe, 2014)	$\Theta(mkn \log(n))$	$\Theta(m \log(n))$	m : number of random trees k : number of features considered for splitting
XGB (Chen and Guestrin, 2016)	$\mathcal{O}(mhn \log(n))$	$\mathcal{O}(mh)$	m : total number of trees h : maximum depth of the tree

Table 6.2
Average training time and runtime for different algorithms

Surrogate	Training time [s] (training size 9128)	Runtime [μ s] (on a single rod)	Runtime Acceleration (FRAPCON runtime / surrogate runtime)
LUT	–	186.910	3.103E + 04
PLS	0.0288	0.102	5.686E + 07
SVR	224.867	128.082	4.528E + 04
*SVR	0.0318	5.636	1.029E + 06
GP	930.980	1778.278	3.262E + 03
*GP	4.233	167.316	3.466E + 04
NN-1layer	8.361	14.333	4.047E + 05
NN-2layer	10.496	16.389	3.539E + 05
NN-3layer	12.351	18.821	3.082E + 05
RF	67.915	95.513	6.072E + 04
XGB	36.077	12.138	4.778E + 05
**Combined surrogate	~63.160	472.277	1.228E + 04

*GP and *SVR are trained on a smaller training set with 300 samples from the space-filling design

**The combined surrogate composes of the best surrogates as shown in Table 6.3 for prediction of all QoIs at the same time.

stress happens at some time during a power cycle. Therefore, the maximum hoop stress covers a wide range, from negative values (typically around -70 MPa) to positive numbers (up to 190 MPa). The relative errors are magnified for small absolute values (close to zero), which explains the large rRMSE values (over 100%) as shown in Fig. 10e. The RMSE and MAE are 15.3 MPa and 12.4 MPa for the best surrogate. It is also worth mentioning that the accurate prediction of high tensile (positive) hoop stress matters more than that of compressive (negative) hoop stress, as peak tensile hoop stress directly correlates to PCI failure risk. The maximum hoop strain is subject to similar uncertainties and reconstruction errors, and the best R^2 is approximately 0.90.

The last two QoIs (failure probabilities due to PCI-SCC or PCI-MPS) are generally harder to predict due to more complicated physics, as damage accumulation is triggered by the maximum hoop stress above certain temperature and burnup-dependent threshold, and the prediction of maximum hoop stress itself is subject to relatively large error. Regarding MAE which is a more physical error metric for the last two QoIs, the averaged prediction error is only 0.022 % and 0.36 % for PCI-SCC and PCI-MPS failure probability when using two-layered NN as surrogate. Noticing that failure probabilities compose of a portion of zeros and small positive values, especially for PCI-SCC where more than 30% of the responses are zeros. A mixed model is developed for PCI-SCC, which adopts a support vector machine (SVM) model for classification of the zero and non-zero instances, succeeded by a two-layered NN for regression of the non-zero probabilities. The MAE for PCI-SCC can thereby be driven down to 0.013%. The predictivity for these two QoIs are therefore taken as acceptable.

LUTs perform as good as NNs for the maximum plenum pressure, and outperform all ML models for the maximum fuel temperature. For the remaining QoIs, ML models achieve better generalization than LUTs. Among the ML algorithms, NNs appear to be the best on this specific three-core test set.

6.2. Runtime analysis

With the goal of speeding up the full-core fuel performance modeling, the computational complexity (especially time complexity) during surrogate construction and prediction need to be quantified. Let n be the total number of training samples and d represents the input dimension (in this work $d \ll n$). The asymptotic worst-case/average-case time complexity³ during training and prediction for different algorithms are summarized in Table 6.1⁴. The training complexity describes the worst-case/average-case runtime to train an algorithm on the training set containing n samples. The effort devoted to data preparation and model selection is not included in the training complexity. The prediction complexity is the runtime consumed for making prediction on a single test point. LUT is a constructed as a 2D matrix for repeated use in this work, hence the training complexity is not applicable to LUT. For NN, the time complexity depends on both the neural network structure and optimization algorithm so only the prediction complexity is estimated.

Despite that the runtime depends on the training size and model structures, the training/prediction efficiency can be compared for this application, as shown in Table 6.2. Note that the “training time” in Table 6.2 stands for the one-time training time, while in reality such training procedure needs to be repeated a lot of times during hyperparameter tuning and cross validation. The overall training time can vary significantly in response to different searching ranges of the hyperparameters and the total number of algorithm fitting, therefore the one-time training time is reported instead. It is also worth mentioning that the runtime of each algorithm listed in Table 6.2 also depends on the implementation efficiency of the python library used. Although LUT does not run the fastest, it has extensive applicability without the need to be reconstructed for new operating conditions. PLS runs the fastest as a projection plus linear regression method. Tree-based algorithms are also fast in prediction, with XGB being more efficient than RF during both training and prediction. NNs are relative slow for both training and prediction, as numerous hyperparameters embedded in NNs increase the computational cost during both model selection and prediction. The training cost for kernel-based methods

³ Big-O or $\mathcal{O}$ is the asymptotic notation for the worst-case runtime. Big Theta or Θ denotes the asymptotically tight bound on the runtime.

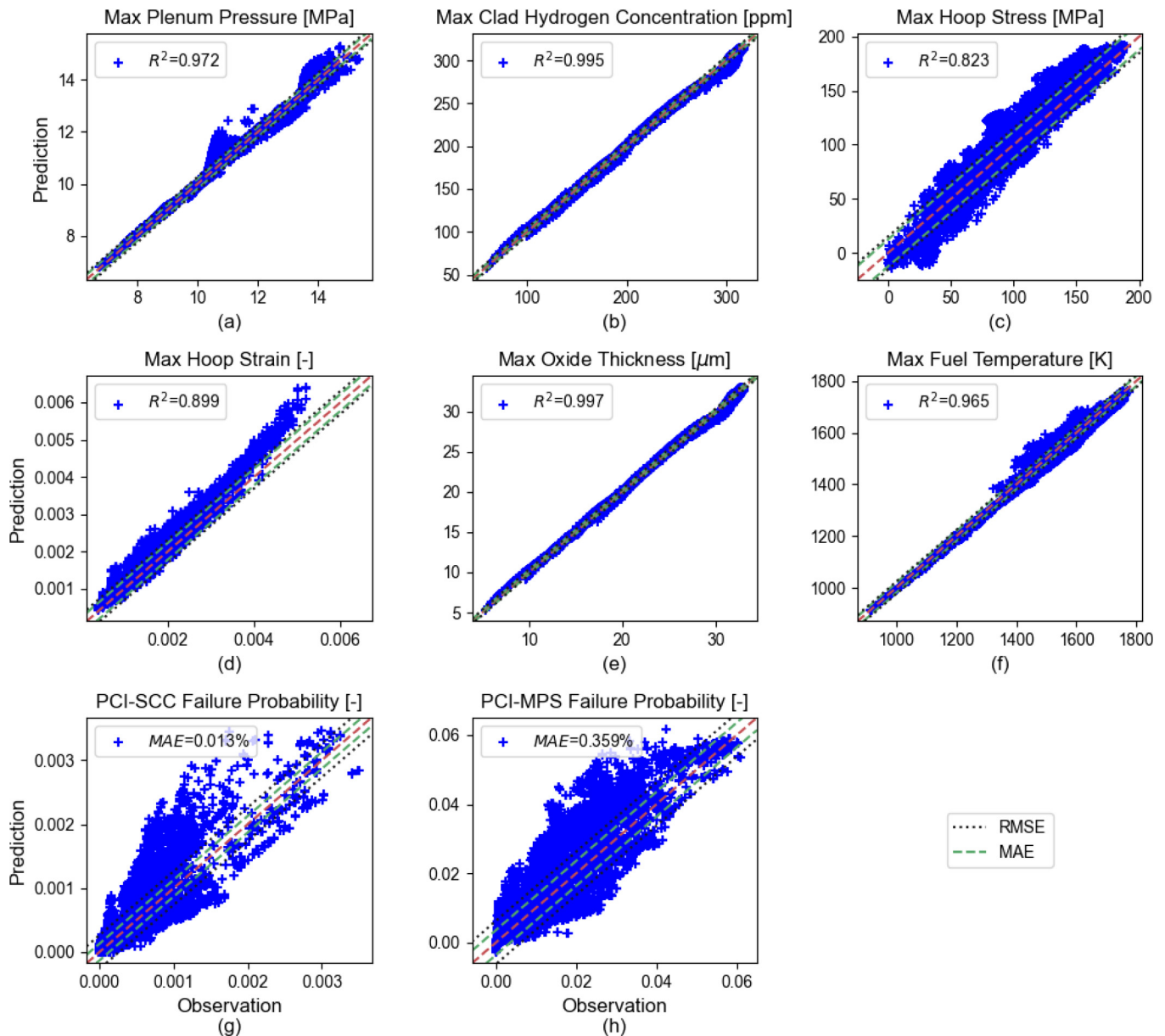
⁴ Note that analytic form of the time complexity depends on specific implementation of the algorithm. The analyses shown in Table 7 are intended for a general comparison.

Table 6.3

Best surrogates of fuel performance QoIs.

QoIs	2-cycle rods		3-cycle rods	
	Best Surrogate	R^2 (*MAE)	Best Surrogate	R^2 (*MAE)
Max Plenum Pressure [MPa]	LUT	0.972	LUT	0.960
Max Hydrogen Concentration [ppm]	NN-2layer	0.995	NN-2layer	0.985
Max Hoop Stress [MPa]	NN-2layer	0.922	NN-3layer	0.904
Max Hoop Strain [-]	NN-2layer	0.899	NN-2layer	0.789
Max Oxide Thickness [μm]	NN-2layer	0.997	NN-2layer	0.98
Max Fuel Temperature [K]	LUT	0.963	LUT	0.912
PCI-SCC Failure Probability [-]	mixed (SVM + NN-2layer)	*0.013%	mixed (SVM + NN-2layer)	*0.0050%
PCI-MPS Failure Probability [-]	NN-2layer	*0.36%	NN-3layer	*0.21%

Note: SVM is a support vector machine classifier trained to distinguish between zero and non-zero failure probabilities.

**Fig. 11.** Predictions versus observations with the best surrogate for each fuel performance QoI (2-cycle rods).

like GP explodes as training size increases, while the prediction accuracy does not improve much (and possibly reduces the prediction accuracy). Therefore, a smaller training set of size 300 is gen-

erated specially for GP and SVR to avoid explosion in computational cost, which effectively reduces both the training and prediction time. When tuning a single-layered NN, a wider

Table 6.4

Design characteristics for two equilibrium PWR cores with extended end-of-life burnup (label number refers to the number of core reload patterns that were evaluated until a strong candidate was found)

Core Label	Blanket Enrichment [%]	Heat Flux Hot Channel Factor (F_Q) [-]	F-delta-H [-]	Boron Concentration [ppm]	Cycle Length [days]	Maximum Burnup [GWD/tU]	Fuel LCOE [\$/MWh]
A	3.2	1.987	1.531	1707	687.5	76.9	5.31
B	3.2	2.022	1.527	1706	689	77.2	5.30
Standard PWR	/	2.0	1.52	1500	500	62	5.67

hidden layer is used to capture the non-linear response, compared to NNs that are deeper but each hidden layer is of smaller width. Therefore, the two-layered NN proves to be the most efficient regarding actual training/prediction time. In comparison to FRAPCON which takes approximately 5.8 s for a single two-cycle rod, the combined surrogate that contains the best surrogate for each QoI (Table 6.3) achieves around 10,000 times acceleration as listed in the last column of Table 6.2. This provides great potential to assist the core design optimization procedure where large number of patterns are need to be investigated.

6.3. Selected surrogate for each fuel performance QoI

Selection of the most desirable surrogate requires comprehensive consideration of prediction accuracy and efficiency. The judge on prediction accuracy prefers not only high predictivity coefficient R^2 , but also low RMSE, MAE and fewer outliers. When the prediction precisions come out comparable for two algorithms, it is more ideal to select the one with simpler structure or the one that runs faster. Particularly, it is always rewarding to employ LUT as surrogate when it is capable to provide sufficient predictive power, as LUT is of more general applicability compared to other ML algorithms. Out of these considerations, the recommended surrogates are shown in Table 6.3. LUT and NNs are generally more preferable. Predictive performance versus true observations with the selected best surrogates for the 2-cycle rods are shown in Fig. 11.

6.4. Application to high burnup cores

The same approach for surrogate construction has also been successfully applied onto equilibrium PWR cores with high burnup, as described briefly in this section. Extending the peak rod average burnup beyond the current regulatory limit (62 GWD/tU) helps extend the fuel cycle operation up to 24 months, reduce

the number of fresh fuel assemblies, therefore improve the economic efficiency of nuclear power plants. With the expected peak rod average burnup reaching 75 GWD/tU, Table 6.4 shows the design characteristics of two representative high burnup cores from neutronics analyses. Assemblies in high burnup cores experience up to three in-core cycles. Certain assemblies are rotated at the beginning of cycle for better neutronics performance. F_Q measures the peak power in the hot channel relative to the average power within the core. F-delta-H describes the ratio of the maximum integrated rod power to the average rod power. By extending the maximum burnup to more than 75 GWD/tU, the fuel leveled cost of electricity (LCOE) can be reduced by $\sim 13\%$ compared to standard PWR core reload designs that were shown in the previous sections. Preliminary full-core fuel performance modeling result for one high burnup core (core-A) is presented in Appendix D in detail. The core reload pattern design and fuel performance is similar to the high burnup core designed by Southern Company (Capps et al., 2020). As shown in Fig. 12 (a) and (b) for core-A, the core-wide maximum plenum pressure can be as high as 24.5 MPa in the design lifetime, well exceeding the coolant pressure (15.5 MPa) for a prolonged time. Such high plenum pressure will lead to high cladding tensile strain as shown in Fig. 12 (c), which results in 645 out 6864 number of rods to exceed above the 1% hoop strain regulatory limit. The same conclusion holds for the other high burnup core-B as presented in Fig. 13. Despite that the neutronics analysis for the two cores are essentially the same, the difference in fuel performance is not negligible. For instance, for core-B, the limiting fuel is neither the one with peak LHGR or peak burnup. Such sensitivity in the fuel performance response further motivates integrating the fuel performance feedback into the reactor core design by constructing the fast-running surrogates.

Following the identical methodology shown in the previous sections, best surrogates are selected for each fuel performance QoI in Table 6.5. Fuels going through extended burnup range exhibits

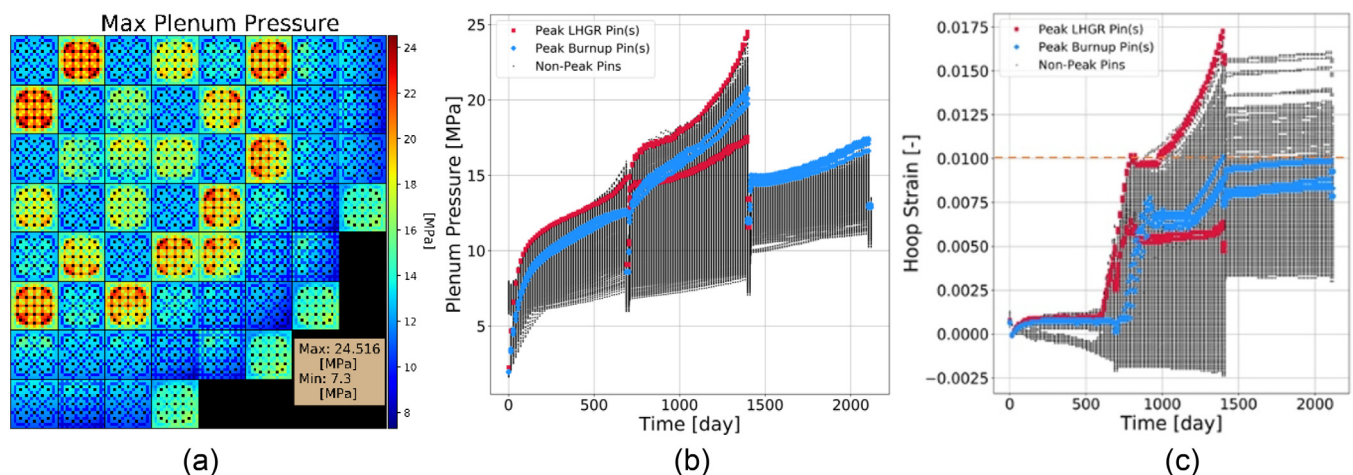


Fig. 12. Full-core fuel performance response of the high burnup core (core-A). (a). Core-wide distribution of the maximum plenum pressure. (b). Evolution of plenum pressure as a function of time. (c). Evolution of cladding hoop strain as a function of time (645 out of 6864 rods are above 1%).

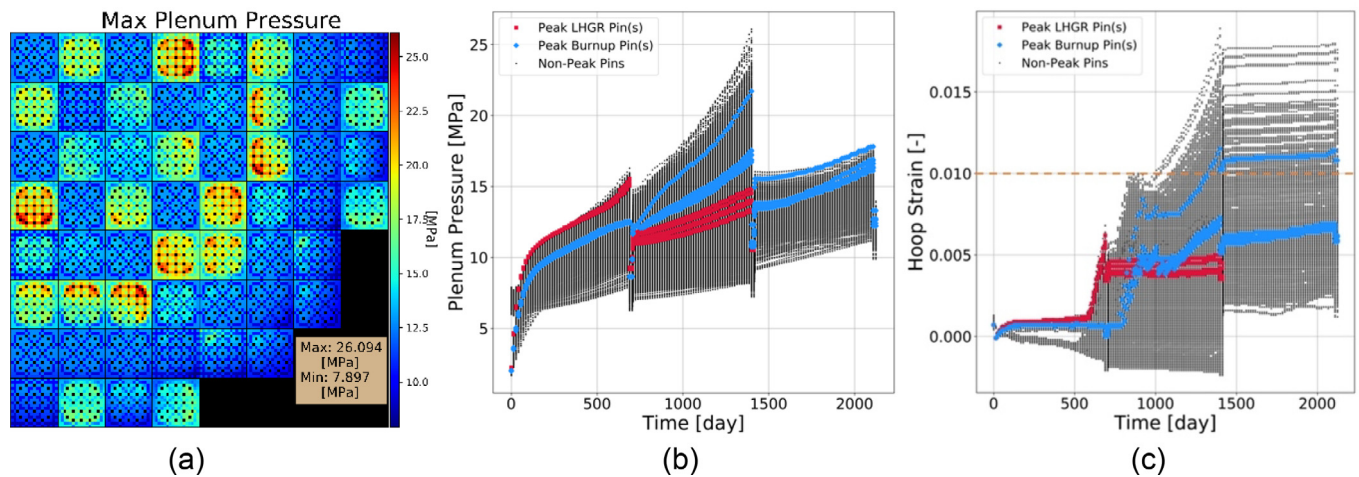


Fig. 13. Full-core fuel performance response of the high burnup core (core-B). (a). Core-wide distribution of the maximum plenum pressure. (b). Evolution of plenum pressure as a function of time. (c). Evolution of cladding hoop strain as a function of time (465 out of 6864 rods are above 1%).

Table 6.5

Best surrogates of fuel performance QoIs for high burnup cores, tested on the two high burnup cores described in Table 6.4.

QoIs	2-cycle rods		3-cycle rods	
	Best Surrogate	R^2 (*MAE)	Best Surrogate	R^2 (*MAE)
Max Plenum Pressure [MPa]	NN-2layer	0.975	NN-2layer	0.968
Max Hydrogen Concentration [ppm]	NN-2layer	0.995	NN-1layer	0.930
Max Hoop Stress [MPa]	NN-2layer	0.844	NN-2layer	0.885
Max Hoop Strain [-]	NN-3layer	0.882	NN-2layer	0.885
Max Oxide Thickness [μm]	NN-3layer	0.995	NN-2layer	0.956
Max Fuel Temperature [K]	LUT	0.900	NN-2layer	0.934
PCI-SCC Failure Probability [-]	mixed (SVM + NN-2layer)	*0.088%	GP	*0.052%
PCI-MPS Failure Probability [-]	mixed (SVM + NN-2layer)	*0.73%	NN-1layer	*1.2%

more complicated thermo-mechanical response. Therefore, ML surrogates with increased model complexity are trained to capture such input/output relation. A runtime analysis shows that it takes 0.0052 s and 0.0041 s respectively to predict all the eight fuel performance QoIs using the ML surrogates for a single 2-cycle or 3-cycle rod. The total runtime acceleration compared to FRAPCON simulation becomes around 1000 and 2000 times as a result. It is also worth pointing out that the ML algorithms offer better applicability and predictivity that generally outperform LUTs for high burnup cores that embeds higher nonlinearity due to more complicated feedbacks.

7. Conclusions

This work provides a first study on constructing fast-running surrogates for essential fuel performance QoIs to accelerate the full-core fuel performance analysis. Constructed surrogates show great predictive accuracy on both unseen standard cores and high burnup cores, and can efficiently reduce the runtime by at least three orders of magnitude compared to fuel performance analysis code FRAPCON. As such, it allows for direct coupling of the fuel performance responses into the core design optimization, especially for high burnup cores and potential advanced reactor types. For certain fuel performance QoIs that embed complicated physics, ML models compensate for the inadequacy of conventional LUTs. Different techniques for feature engineering and experimental design are introduced and used, depending on the required applicability of the surrogate. The best surrogate can be thereby selected based on overall consideration of both prediction accuracy and time complexity. A comprehensive

study is provided to demonstrate the application and applicability of LUTs versus different ML models for each fuel performance QoI, and the following conclusions can be drawn from this work:

- As a feature extraction technique, rule-based model offers enough flexibility hence extensibility to unseen data range.
- In order to extend the applicability of ML algorithms to unobserved core patterns, an extended space-filling design is performed on the enlarged domain to make sure that the training data sufficiently covers the data range with physical desirability.
- With operating power being the only varying condition, the maximum fuel temperature and maximum oxide thickness are highly linear with respect to burnup and power level. Therefore LUTs show superiority in both accuracy and simplicity for these two QoIs.
- For other fuel performance QoIs that involve more complicated physics, ML-assisted surrogates outperform LUTs in prediction accuracy.
- NNs are powerful to capture the nonlinear input/output relation and are more robust to noise (reconstruction error) compared to other ML algorithms, therefore favored in the current analysis.
- Regarding the time complexity of the surrogates, the cost of training a ML surrogate reduces in the order of kernel methods (GP and SVR), tree-based algorithms (RF and XGB), neural networks and linear regression (PLS) in a general way. Kernel methods are more suitable to be applied onto smaller training size given their high computational cost, and they tend to be more accurate than other methods trained on sim-

ilar sample size. In addition, when the dataset presents relatively large random noise (reconstruction errors in this work), using too dense training samples can harm the performance of kernel methods. Linear regression (PLS) is not suitable for most of the multiphysics problems, but provides a good benchmark.

- For high burnup fuel cores, meeting the neutronic licensing constraints in core reload analysis no longer implies meeting fuel performance licensing parameters (e.g. cladding hoop strain limit). Fast-running predictive surrogates could be essential part of core optimization process for high burnup fuels.

Beyond the study conducted in this paper, potential efforts should be made in the future work:

- Updating physical range for the extended space-filling design, upon availability of more fuel loading patterns.
- Increasing the model complexity by taking into consideration more physical parameters, e.g. number of batches, power cycle length, load following etc. Inclusion of such scalar variables likely increases the input dimension without adding extra burden to the overall workflow.
- Generalizing the full-core surrogates to high-fidelity fuel performance code like BISON for more accurate fuel integrity monitoring.
- Integrating the fast-running surrogates with core reload optimization to achieve tighter coupling of fuel performance feedbacks to other physics.

CRedit authorship contribution statement

Yifeng Che: Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Joseph Yurko:** Methodology, Writing – review & editing. **Paul Seurin:** Investigation. **Korosh Shirvan:** Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Calculation of rod failure risk

Under the low temperature steady-state operating condition, a cumulative damage model (Rashid et al., 1989; Hales et al., 2016) can be used to estimate the cladding damage induced by stress corrosion cracking during PCI. The amount of damage (cumulative damage index, CDI) at time t_n is defined by

$$CDI = \int_0^{t_n} \frac{dt}{t_f(\sigma, Bu, T)}$$

where t_f is the time-to-failure (in seconds) as a function of the clad hoop stress σ (MPa), burnup Bu (MWd/MTU) and temperature T (K). The time-to-failure t_f is calculated via

$$t_f = \bar{t} \exp\left[(1.015\sigma_y + 1.74\sigma_{ref} - 2.755\sigma) \times 10^{-2}\right]$$

where

$$\bar{t} = 5 \times 10^5 \left(1.13 \times 10^{-4} Bu - 0.13\right)^{-0.75} \exp(-30(1 - 611/T))$$

σ_{ref} is a threshold stress (MPa) in form:

$$\sigma_{ref} = \begin{cases} 336.476(Bu - 5000)^{-0.07262} Zr2 \\ 310.275(Bu - 5000)^{-0.04400} Zr4 \end{cases}$$

Cladding yield stress σ_y (MPa) as a function of cladding temperature is interpolated according to (Cazalis et al., 2007). The CDI model is only activated when hoop stress exceeds the threshold stress, and burnup greater than 5000 MWd/MTU. The damage accumulation time is limited to 1000 seconds upon reaching the peak hoop stress based on in-reactor PCI observations, as suggested in (Joseph and Rashid, 2020).

To account for the local stresses and keep consistent with the developed failure metrics, the 2D radial-axial (R-Z) hoop stress is mapped to radial-azimuthal (R- θ) through stress concentration factors (CFs) (Joseph and Rashid, 2020), estimated as:

$$CF_{PCI-SCC} = -0.0042\sigma_\theta + 2.3773$$

$$CF_{PCI-MPS} = -0.0115\sigma_\theta + 4.3099$$

The cumulative distribution function for the cumulative damage index is described as below. The rod failure risk can be statistically determined from the cumulative distribution function, and rods with failure probability greater than 0.5 are vulnerable to PCI failures. The procedure to calculate the rod failure risk is summarized in Algorithm A.1.

Algorithm A.1: Fuel rod failure risk calculation

- 1: Perform fuel performance analysis and identify the peak R-Z hoop stress σ_θ .
- 2: Apply the stress concentration factor for PCI-SCC and PCI-MPS to the R-Z hoop stress σ_θ .
- 3: Calculate CDI for PCI-SCC and PCI-MPS.
- 4: Infer the rod failure risk from the cumulative distribution function of CDI according to Fig. A.1.

Appendix B Experimental design with extended space-filling design

The extended space-filling design is used to create the synthetic training set to predict unseen core patterns. Fig. B.1(a) to (d) show the procedure step by step based on three available equilibrium

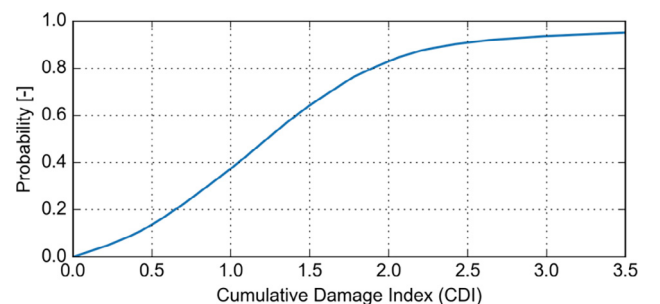


Fig. A1. Cumulative distribution function of CDI.

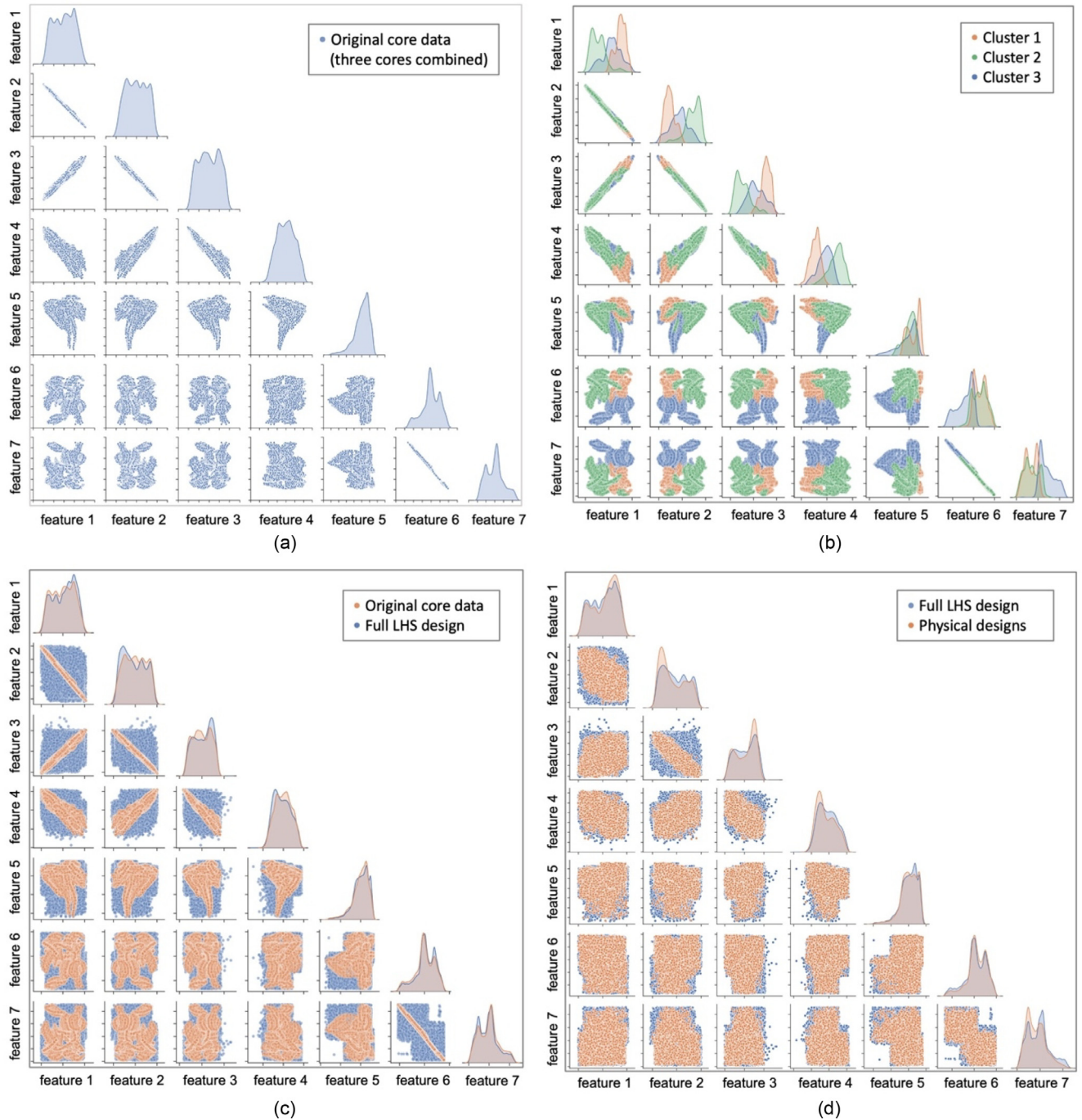


Fig. B1. Extended space-filling design on the rule-based model feature space. (a). pairwise distribution of the features for three available cores. (b). clustering of the features. (c). LHS samples generated according to the marginal probability distributions within each cluster. (d). physical samples after ruling out the unphysical samples. Only the first seven features are shown for simplicity.

PWR cores. First, the rule-based model (polynomial representation) is used for feature extraction, with the pairwise distribution in the feature space shown in Fig. B.1 (a) for only the first seven features for better visibility. The feature space is then divided into local regions by clustering (Fig. B.1 (b)). Fig. B.1 (c) plots the LHS samples generated according to the fitted marginal distributions,

with the original core data visualized on top of the LHS samples. The LHS samples retain the same data range as the original core data with nice space-filling property in the probability space. Finally, LHS samples that correspond to unphysical operating conditions are discarded, and Fig. B.1 (d) presents the physical samples versus the full LHS design. In the current study, 16,632 two-cycle

Table C1

Predictive performance of LUT vs. ML algorithms for 2-cycle rods (green denotes the best desirability, and blue indicates the worst performance)

		Max Plenum Pressure [MPa]	Max Hydrogen Concentration [ppm]	Max Hoop Stress [MPa]	Max Hoop Strain [-]	Max Oxide Thickness [μm]	Max Fuel Temperature [K]	PCI-SCC Failure Probability [-]	PCI-MPS Failure Probability [-]
R^2	LUT	0.972	0.738	-0.044	-5.353	0.742	0.963	-476.882	-0.385
	PLS	0.053	0.946	0.682	-4.541	0.946	0.542	-1.609	-0.214
	SVR	-0.676	0.971	0.755	-5.876	0.971	0.484	-0.905	0.285
	*SVR	0.974	0.983	0.786	0.727	0.984	0.929	0.138	0.285
	GP	0.400	0.965	0.721	-3.031	0.964	0.584	-0.845	0.243
	*GP	0.985	0.991	0.856	0.740	0.991	0.924	0.112	0.391
	NN-1layer	0.885	0.952	0.882	0.587	0.866	0.888	0.366	0.625
	NN-2layer	0.986	0.995	0.922	0.899	0.997	0.907	0.374	0.720
	NN-3layer	0.980	0.995	0.901	0.842	0.993	0.955	0.486	0.844
	RF	0.768	0.754	0.761	-1.768	0.750	0.775	-0.507	0.534
RMSE	XGB	0.441	0.857	0.736	-1.973	0.872	0.735	-0.716	0.480
	LUT	0.263	35.332	53.927	2.322E-03	3.721	22.435	9.283E-03	1.649E-02
	PLS	1.545	14.443	30.976	2.298E-03	1.527	86.221	7.214E-04	1.304E-02
	SVR	2.055	10.487	27.168	2.560E-03	1.115	91.580	6.164E-04	1.001E-02
	*SVR	0.258	8.063	25.425	5.105E-04	0.831	34.034	4.145E-04	1.001E-02
	GP	1.229	11.646	29.013	1.960E-03	1.250	82.187	6.065E-04	1.030E-02
	*GP	0.197	5.979	20.824	4.980E-04	0.636	35.070	4.208E-04	9.239E-03
	NN-1layer	0.538	13.589	18.822	6.273E-04	2.419	42.638	3.556E-04	7.251E-03
	NN-2layer	0.187	4.521	15.312	3.099E-04	0.334	38.920	3.534E-04	6.259E-03
	NN-3layer	0.225	4.330	17.253	3.885E-04	0.534	27.091	3.201E-04	4.670E-03
MAE	RF	0.764	30.753	26.863	1.624E-03	3.295	60.417	5.482E-04	8.080E-03
	XGB	1.187	23.421	28.222	1.683E-03	2.365	65.625	5.850E-04	8.537E-03
	LUT	0.194	29.684	45.753	1.759E-03	3.132	17.684	7.112E-03	1.268E-02
	PLS	1.421	11.964	25.305	2.096E-03	1.264	75.035	5.974E-04	1.094E-02
	SVR	1.518	8.123	22.181	2.084E-03	0.860	66.285	3.862E-04	7.129E-03
	*SVR	0.165	6.062	20.053	3.318E-04	0.625	26.809	2.521E-04	6.397E-03
	GP	1.009	9.000	23.205	1.623E-03	0.966	60.676	4.029E-04	7.600E-03
	*GP	0.124	3.956	15.946	3.073E-04	0.420	27.647	2.411E-04	5.783E-03
	NN-1layer	0.399	10.357	15.074	4.368E-04	1.858	35.847	2.356E-04	5.158E-03
	NN-2layer	0.140	3.292	12.381	2.251E-04	0.259	28.179	2.416E-04	3.593E-03
Max Error	NN-3layer	0.165	3.442	13.921	2.675E-04	0.441	21.634	2.159E-04	2.770E-03
	RF	0.578	27.270	20.831	1.158E-03	2.902	41.211	3.160E-04	5.399E-03
	XGB	0.888	19.492	22.024	1.225E-03	1.930	44.872	3.170E-04	5.653E-03
	LUT	1.391	71.779	132.727	4.490E-03	7.533	105.551	2.138E-02	5.251E-02
	PLS	9.103	269.837	290.055	5.381E-03	28.877	867.291	1.890E-03	3.632E-02
	SVR	11.965	31.809	78.506	8.506E-03	2.636	592.881	2.714E-03	3.847E-02
	*SVR	1.910	33.910	85.583	2.035E-03	3.222	160.684	2.560E-03	5.904E-02
	GP	3.736	48.060	84.890	5.491E-03	3.635	587.723	2.797E-03	4.046E-02
	*GP	1.597	33.272	97.969	1.763E-03	3.633	158.824	2.598E-03	4.717E-02
	NN-1layer	7.318	283.558	260.418	3.339E-03	30.316	806.789	2.555E-03	5.071E-02
Min Error	NN-2layer	8.158	257.293	272.003	6.019E-03	27.727	851.586	3.658E-03	6.186E-02
	NN-3layer	8.750	255.192	271.572	6.123E-03	27.313	840.176	3.323E-03	5.503E-02
	RF	4.213	62.421	99.309	6.418E-03	6.814	436.049	2.724E-03	3.633E-02
	XGB	12.198	255.521	288.056	8.091E-03	28.199	890.304	3.569E-03	5.214E-02
	LUT	-0.726	-80.039	-65.362	-3.337E-04	-8.379	-53.778	-3.048E-05	-4.025E-02
	PLS	-9.631	-247.385	-358.316	-8.859E-03	-26.308	-673.366	-5.995E-03	-1.025E-01
	SVR	-3.623	-83.015	-110.200	-4.037E-03	-8.817	-148.101	-2.211E-03	-3.374E-02
	*SVR	-1.862	-32.488	-80.613	-2.808E-03	-3.507	-125.608	-1.462E-03	-3.383E-02
	GP	-1.923	-54.736	-60.597	-1.516E-03	-6.649	-200.431	-1.799E-03	-3.153E-02
	*GP	-1.368	-30.751	-82.754	-3.045E-03	-3.248	-166.085	-1.177E-03	-2.979E-02
	NN-1layer	-8.631	-306.946	-333.554	-4.801E-03	-40.756	-829.239	-3.870E-03	-6.483E-02
	NN-2layer	-7.603	-261.059	-277.970	-4.779E-03	-27.745	-713.290	-3.503E-03	-6.373E-02
	NN-3layer	-7.711	-268.349	-261.718	-4.828E-03	-27.204	-785.691	-3.463E-03	-6.243E-02
	RF	-2.013	-70.009	-71.960	-3.148E-03	-8.094	-147.000	-1.566E-03	-3.144E-02
	XGB	-7.948	-233.524	-265.454	-4.932E-03	-25.592	-591.569	-3.644E-03	-6.337E-02

rods are available from the three PWR cores. The feature space is divided into three clusters, and 8,000 samples are generated within each region. Upon screening out the unphysical samples, a total number of 9,128 samples are left and used as training set for the ML algorithms.

Appendix C Predictive performance of LUT and ML surrogates in details

Detailed comparison of the surrogate performance is summarized for 2-cycle rods in Table C.1, including the predictivity

coefficient R^2 , RMSE, MAE, maximum and minimum prediction errors on the test set (three unseen cores). Fig. C.1 shows the surrogate prediction errors versus reconstruction errors with the selected best surrogates for 2-cycle rods. A better way to understand the relationship between the surrogate prediction accuracy and reconstruction errors is shown in Fig. C.2, where the absolute prediction errors are plotted in boxplots for different absolute reconstruction error quantiles. It is reasonable that the prediction errors can be relatively large for rods subject to great reconstruction errors, and the boxplot medians should readily increase with respect to the reconstruction errors. Separated by IFBA rods and regular rods, it is indeed the case for all the QoIs, except for the

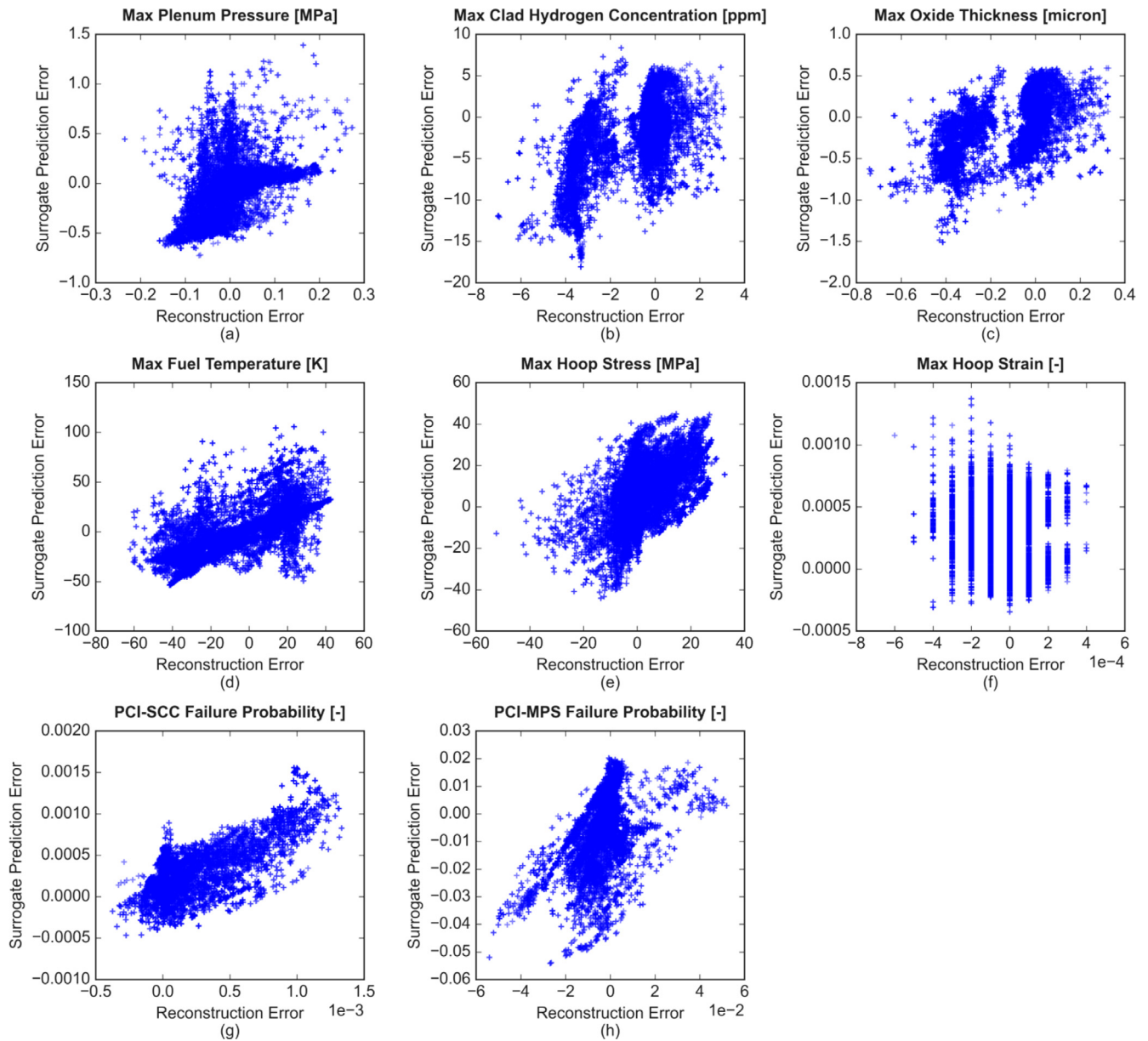


Fig. C1. Surrogate prediction error versus reconstruction error with the selected best surrogate for the eight fuel performance QoIs for 2-cycle rods.

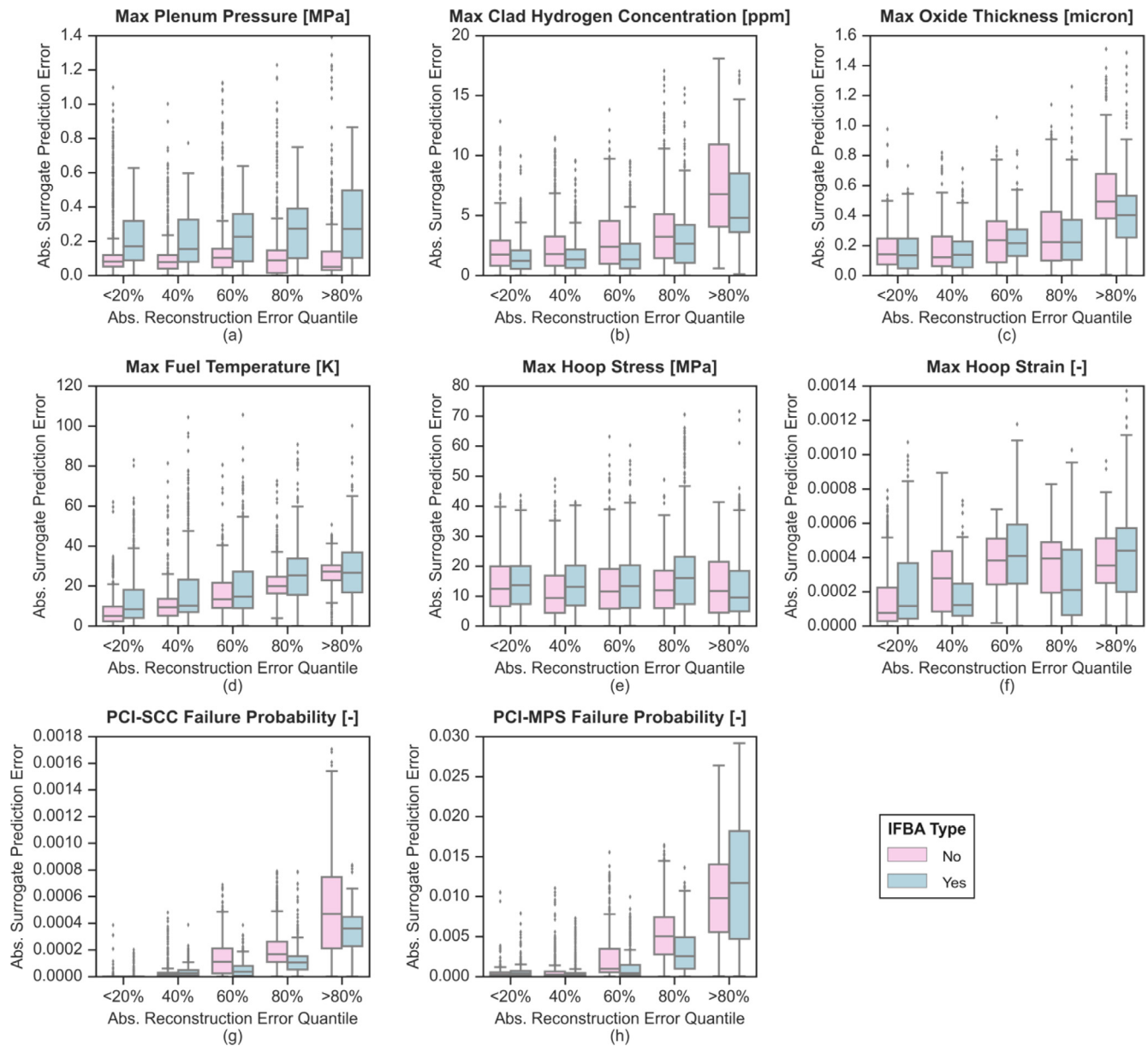


Fig. C2. Absolute prediction errors in boxplots for different absolute reconstruction error quantiles, with the selected best surrogate for the eight fuel performance QoIs for 2-cycle rods. Pink and blue boxplots are for regular and IFBA rods respectively.

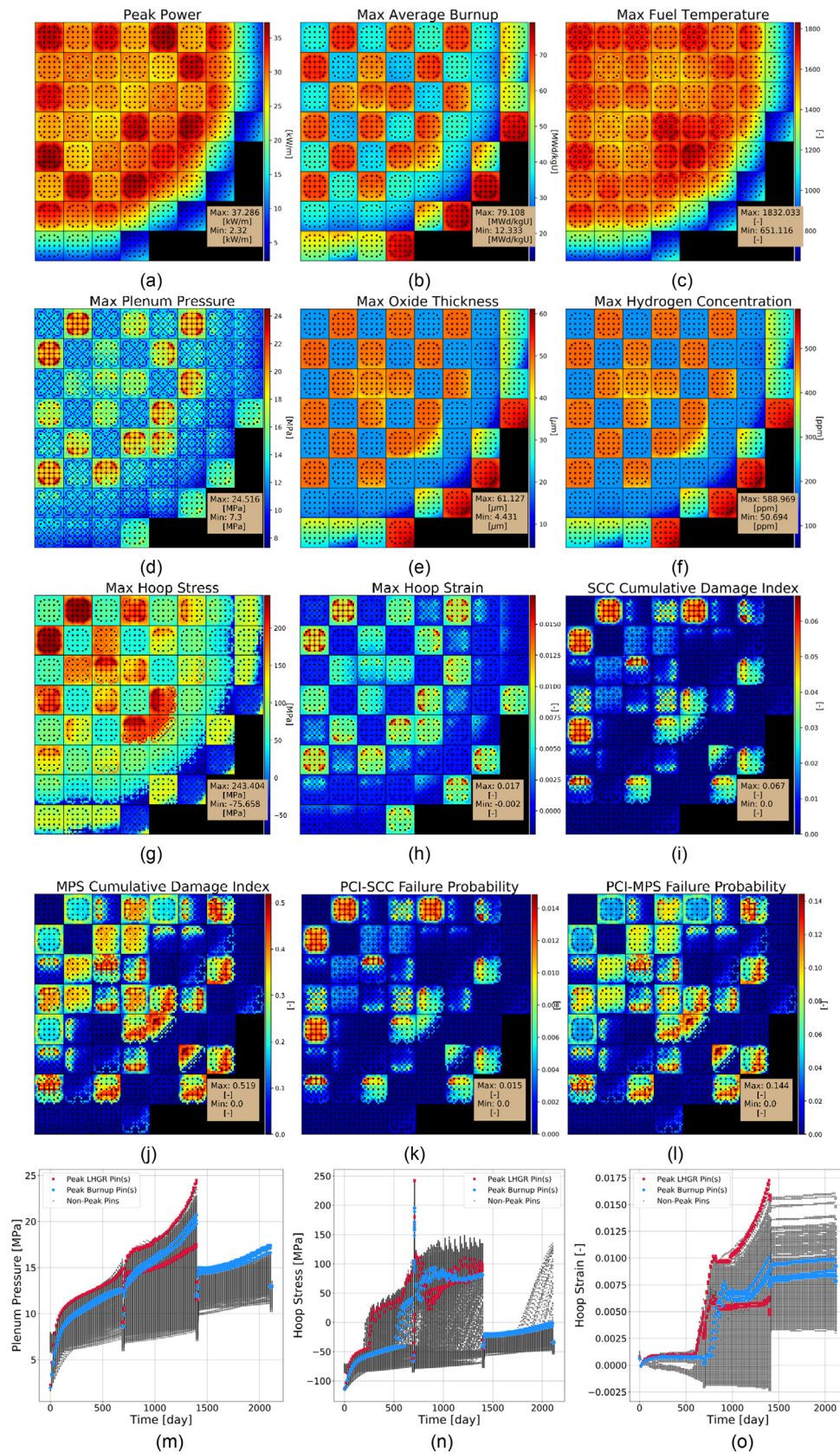


Fig. D1. Full-core fuel performance modeling result for core-A. Subfigures (a) to (i) exhibits the core-wide distribution of the maximum values of quantities of interest during the entire reactor operation. Subplots (m) to (o) show the evolution of maximum fuel temperature, plenum pressure and maximum hoop stress as a function of time. Fuel pins with peak linear power are marked with red squares, and the blue diamonds mark the most burnt fuel pins.

maximum hoop stress which is sensitive to the operating condition and subject to large reconstruction errors.

Appendix D Preliminary full-core fuel performance modeling result for a representative high burnup core design

Full-core fuel performance modeling has been performed for high burnup core designs from preliminary neutronics analysis. Fig. D.1 (a) to (i) show the full-core result in core-wide distribution of the maximum values of QoIs throughout the entire life, and (m) to (o) present the evolution of QoIs as a function of time for core-A. Using the existing fuel performance modeling tools, fuel rods undergo more severe challenges, i.e., higher plenum pressure, fuel temperature, and cladding hoop strain. Fuel performance QoIs exceeding the regulatory limits will require redesign (e.g. shorter stack fuel to allow more plenum space or moving to lower burnup). This preliminary study implies the requirement for better core designs of fuel loading patterns.

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